

A Comparative Study of Multivariate Statistical Techniques And Machine Learning Algorithms To Predict The Survival Of Startup Firms

PROJECT

Submitted in Partial Fulfillment of the Requirements

For the Award of the Degree of

MASTER OF SCIENCE

IN

STATISTICS

By

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MARCH 2026

PRESIDENCY COLLEGE (AUTONOMOUS), CHENNAI - 05

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DECLARATION

I hereby declare that the project titled "**A Comparative Study of Multivariate Statistical Techniques And Machine Learning Algorithms To Predict The Survival Of Startup Firms**" submitted in partial fulfilment for the requirements for the award of Master's degree in Science in Statistics record of original project work done by me under the supervision and Guidance of **Dr. R. SAKTHIVEL**, Associate Professor, Department of Statistics, Presidency college (Autonomous), Chennai-05. This Project has not been submitted for the award of any other degree in any other University/Institute previously.

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CERTIFICATE

This is to certify that the project titled "**A Comparative Study of Multivariate Statistical Techniques And Machine Learning Algorithms To Predict The Survival Of Startup Firms**" submitted by **SARANKUMAR K, (Reg.No: 2413212075007)** in partial fulfillment for the award of Master's Degree in Science in Statistics is a record of original work done by her under my Supervision and Guidance. This work has not formed the basis for the award of any degree or diploma previously.

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This report is submitted to the department of statistics for Viva – Voce held on March 2026

Examiners

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Dr. R. RAMAN**, Joint director(F) and Principal (FAC), Presidency College (Autonomous), Chennai, for providing me with the opportunity and facilities to complete this project successfully.

I am extremely thankful to **Dr. N. VISWANATHAN**, Associate and Head of the Department of Statistics, Presidency College (Autonomous), Chennai-05, for his valuable support, encouragement, and guidance throughout the course of this project.

I would like to express my deepest gratitude to my project Guide **Dr. R. SAKTHIVEL**, Associate Professor, Department of Statistics, Presidency College (Autonomous), Chennai-05 for the invaluable guidance, encouragement, and constructive suggestions which helped me successfully complete this project. Their expertise and dedication have inspired me to strive for excellence and have profoundly shaped my academic and personal growth.

Lastly, I extend my sincere thanks to everyone who, directly or indirectly, contributed to the completion of this work. Their encouragement, assistance, and cooperation have been invaluable.

SARANKUMAR K

CONTENTS

CHAPTER	TITLE	PAGE NO
1	INTRODUCTION	1
2	AIMS AND OBJECTIVES	15
3	REVIEW OF LITERATURE	18
4	METHODOLOGY	25
5	RESULTS AND DISCUSSION	46
6	CONCLUSION AND SUMMARY	96
	REFERENCE	100

CHAPTER I

INTRODUCTION TO STARTUP COMPANIES

1.1 Introduction



In the contemporary global economy, startup companies have emerged as one of the most dynamic and transformative forms of enterprise organization. The shift from industrial economies to knowledge-driven economies has elevated the importance of innovation, creativity, and technological advancement. Within this context, startups represent structured attempts to convert ideas into scalable economic value under conditions of uncertainty. They are not merely small businesses in early stages; rather, they are innovation-oriented ventures designed for rapid growth and long-term structural impact.

The prominence of startups has expanded significantly over the past three decades. Rapid technological advancement, digital infrastructure development, globalization of markets, and the growth of venture capital networks have collectively reduced entry barriers for new ventures. Entrepreneurs today can conceptualize, develop, and launch innovative products or services with comparatively lower initial investment than in traditional industrial sectors. As a result, startups have become critical engines of employment generation, technological progress, and competitive dynamism.

Startups operate in uncertain environments characterized by evolving consumer preferences, technological volatility, regulatory complexity, and intense competition. Despite these challenges, their potential for exponential growth distinguishes them from traditional enterprises. The capacity to scale rapidly, especially through digital platforms and network effects, positions startups as central contributors to economic modernization.

This chapter provides a comprehensive examination of startup companies from conceptual, structural, historical, and strategic perspectives. It aims to establish a strong theoretical foundation for understanding startups as distinct organizational entities within modern economies.

1.2 Concept and Definition of Startup Companies

A startup company can be defined as a newly established organization formed to develop innovative products, services, or processes, operating under conditions of high uncertainty and structured for scalable growth. This definition emphasizes three essential elements: innovation, uncertainty, and scalability.

Innovation forms the core of startup identity. Unlike conventional businesses that replicate established models, startups introduce new value propositions. Innovation may be technological, process-oriented, service-based, or model-driven. The essential criterion is differentiation from existing market offerings.

Uncertainty refers to the unpredictable environment in which startups function. Market demand may not yet be validated. Regulatory frameworks may evolve. Technological feasibility may require continuous refinement. Startups therefore engage in iterative experimentation to reduce uncertainty over time.

Scalability distinguishes startups from small enterprises. A scalable business model enables expansion without proportional increases in cost. Digital platforms, automation, and global connectivity allow startups to reach large customer bases with relatively limited physical infrastructure.

It is essential to differentiate startups from traditional small businesses. A small retail store, for instance, may be newly established but operates within a proven model aimed at stable income generation. A startup, in contrast, is structured to grow rapidly and potentially transform industry standards.

1.3 Historical Evolution of Startup Companies

The origins of modern startup culture can be traced to technological transformations of the late twentieth century. The expansion of personal computing, internet connectivity, and software development created opportunities for new forms of enterprise.

The rise of innovation hubs such as Silicon Valley played a crucial role in shaping startup ecosystems. Supported by academic institutions like Stanford University, venture capital networks, and a culture that tolerated risk and failure, Silicon Valley became a model for innovation-driven entrepreneurship.

Several globally influential corporations originated as startup ventures. Companies such as Amazon, Google, and Facebook began as small experimental enterprises before evolving into multinational corporations. Their trajectories illustrate the transformative potential of startups when innovation aligns with scalability.

Over time, the startup model expanded globally. Cities in Europe, Asia, and the Middle East developed innovation clusters supported by government policy, private investment, and academic collaboration. The globalization of startup culture reflects the universal relevance of entrepreneurship in knowledge-based economies.

1.4 Characteristics of Startup Companies

Startup companies exhibit several distinctive characteristics that differentiate them from traditional enterprises.

1.4.1 Innovation Orientation

Innovation is central to startup identity. Startups seek to solve problems in novel ways, often leveraging emerging technologies. Innovation may involve product development, service enhancement, digital platforms, or new revenue models.

1.4.2 Lean Organizational Structure

Startups typically operate with small teams and minimal hierarchy. Decision-making processes are rapid, and roles are flexible. This structure promotes adaptability and collaboration.

1.4.3 High Risk and Uncertainty

The majority of startups operate without guaranteed market acceptance. Financial instability, technological challenges, and competitive pressures contribute to high failure rates. However, the potential rewards of success justify risk-taking.

1.4.4 Growth-Centric Strategy

Unlike established firms that prioritize stability, startups emphasize expansion. Early-stage strategies often focus on customer acquisition and market penetration rather than immediate profitability.

1.4.5 Scalability and Network Effects

Many startups rely on digital platforms that benefit from network effects, where the value of the product increases as more users participate. This dynamic accelerates growth.

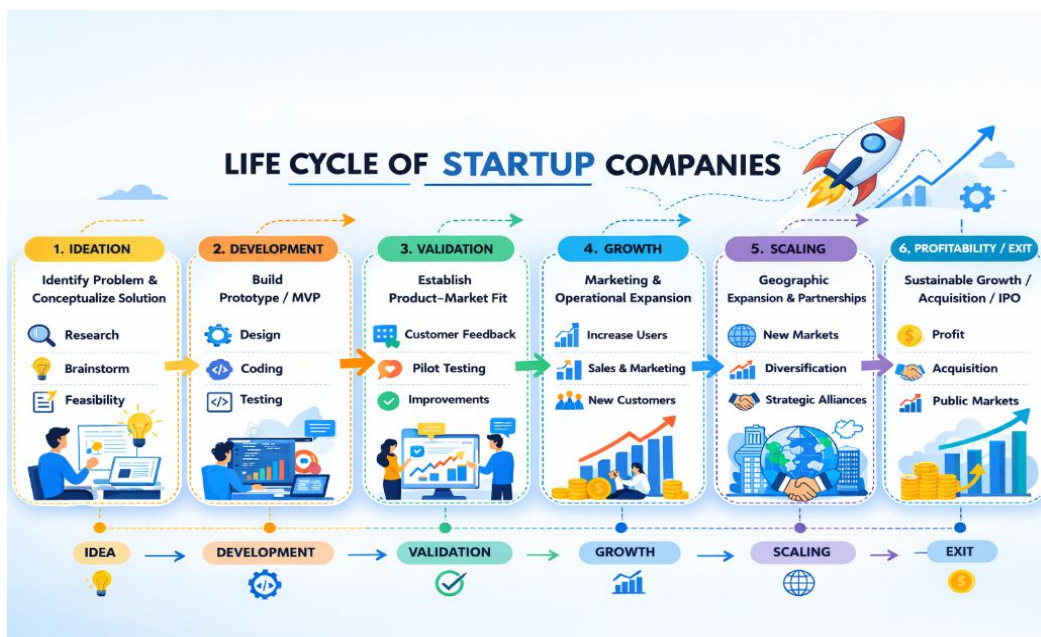
1.5 Types of Startup Companies

Startup companies can be categorized into various types based on their objectives and operational domains.

Technology startups focus on software development, artificial intelligence, cybersecurity, or biotechnology. Platform-based startups operate digital marketplaces connecting users and service providers. Social startups aim to address societal challenges through sustainable business models. Manufacturing startups develop innovative hardware or engineering solutions. Service-based startups redesign traditional services through digital integration.

Each category contributes uniquely to economic diversification and industrial development.

1.6 Lifecycle of Startup Companies



The startup journey typically progresses through several stages.

- **The ideation stage** involves identifying a problem and conceptualizing a solution. Founders conduct preliminary research to assess feasibility.
- **During the development stage**, prototypes or minimum viable products are created to test assumptions.
- **The validation stage** focuses on establishing product-market fit through customer feedback.
- **In the growth stage**, marketing strategies and operational expansion increase customer base.
- **The scaling stage** involves geographic expansion, diversification, and strategic partnerships.

Finally, startups may achieve sustainable profitability, pursue acquisition, or enter public markets.

1.7 Startup Ecosystem

Startups operate within interconnected ecosystems comprising entrepreneurs, investors, universities, incubators, accelerators, government agencies, and customers.

Investors provide financial capital and mentorship. Universities contribute research and skilled graduates. Incubators offer infrastructure and guidance during early stages. Governments establish regulatory frameworks and provide incentives. Customers validate value propositions and generate revenue streams.

The strength of interactions among these stakeholders determines ecosystem vitality.

1.8 Funding Mechanisms of Startup Companies

Access to capital is one of the most critical determinants of startup formation and growth. Unlike traditional firms that may rely primarily on bank loans or internally generated revenue, startups frequently depend on diversified funding sources to support innovation and scaling activities.

The earliest stage of financing is commonly referred to as bootstrapping. During this phase, founders utilize personal savings or contributions from family and friends to initiate operations. Bootstrapping encourages financial discipline and resource optimization; however, it may limit growth potential due to capital constraints.

Angel investors represent the next stage of financial support. These individuals typically invest in early-stage startups in exchange for equity ownership. Beyond financial contribution, angel investors often provide mentorship, industry knowledge, and strategic connections.

Venture capital financing plays a transformative role in high-growth startups. Venture capital firms invest substantial amounts of capital in exchange for equity stakes, anticipating significant returns through scaling and eventual exit. Venture capital funding enables startups to accelerate product development, marketing efforts, and geographic expansion.

Later-stage funding may involve private equity or growth capital, supporting large-scale expansion. Exit mechanisms include acquisition by established corporations or public listing through stock exchanges.

The funding architecture of startups reflects their growth-oriented design. External investment not only supplies capital but also legitimizes the venture within the broader ecosystem.

1.9 Role of Innovation in Startup Success

Innovation serves as the foundational driver of startup competitiveness. The capacity to introduce differentiated solutions determines market entry success and long-term sustainability.

Innovation within startups often follows an iterative model. Rather than developing comprehensive products prior to market entry, startups release minimum viable products to gather user feedback. Continuous refinement reduces uncertainty and enhances product-market alignment.

Technological innovation is particularly prominent in digital startups. Artificial intelligence, cloud computing, blockchain technology, and renewable energy solutions have expanded the scope of entrepreneurial possibilities. However, innovation is not confined to technology; business model innovation—such as subscription-based revenue systems or platform ecosystems—has also redefined industry structures.

Sustained innovation requires adaptive organizational culture. Startups cultivate environments that encourage experimentation, collaboration, and rapid problem-solving.

1.10 Entrepreneurial Mindset and Leadership in Startup Companies

The success of startup companies is closely associated with the mindset and leadership capabilities of their founders. Unlike managers in established corporations who operate within defined structures and procedures, startup leaders function in environments characterized by ambiguity and rapid change. Their ability to navigate uncertainty, inspire teams, and make strategic decisions under incomplete information becomes a decisive factor in organizational survival.

The entrepreneurial mindset encompasses opportunity recognition, resilience, adaptability, and long-term vision. Founders must identify emerging trends, anticipate market needs, and convert abstract ideas into actionable strategies. This process requires both creative thinking and analytical judgment.

Leadership in startups tends to be participatory rather than hierarchical. Small teams encourage open communication and collaborative problem-solving. Founders often engage directly in operational tasks, product development, and customer interaction. Such involvement fosters strong organizational culture and shared purpose.

As startups grow, leadership challenges evolve. Founders must transition from hands-on management to strategic oversight. Delegation, team building, and governance structures become increasingly important. The ability to scale leadership competencies alongside organizational growth determines long-term sustainability.

1.11 Organizational Culture in Startup Companies

Organizational culture within startups plays a pivotal role in shaping performance and innovation outcomes. Culture influences decision-making processes, employee motivation, and the capacity to adapt to change.

Startup culture is typically characterized by experimentation, flexibility, and tolerance for failure. Employees are encouraged to propose ideas, challenge assumptions, and engage in creative problem-solving. This environment stimulates innovation and accelerates product development cycles.

Unlike rigid corporate structures, startups emphasize autonomy and accountability. Team members often assume multiple roles, enhancing

cross-functional collaboration. Informal communication channels reduce bureaucratic barriers and enable rapid feedback loops.

However, maintaining a cohesive culture becomes challenging as startups expand. Rapid hiring and geographic diversification may dilute shared values. Founders must therefore institutionalize cultural principles without compromising agility.

1.12 Strategic Planning in Startup Companies

Strategic planning within startups differs fundamentally from traditional long-term corporate planning. In uncertain environments, rigid multi-year plans may prove ineffective. Instead, startups adopt flexible strategic frameworks that allow continuous revision.

Early-stage strategy focuses on achieving product-market fit. Startups test hypotheses regarding customer needs, pricing models, and distribution channels. Feedback-driven iteration replaces static planning.

As validation strengthens, strategy shifts toward scaling operations. Market segmentation, competitive positioning, and partnership development gain importance. Startups must allocate resources efficiently while maintaining innovation momentum.

Competitive strategy often involves differentiation rather than direct confrontation with established firms. By targeting underserved market segments or introducing cost-effective solutions, startups create niches that reduce competitive pressure.

Strategic agility—the capacity to adapt plans based on evolving information—remains essential throughout the lifecycle.

1.13 Risk Management in Startup Companies

Risk is inherent in startup ventures. Financial instability, technological feasibility challenges, market unpredictability, and regulatory shifts contribute to uncertainty.

Effective risk management does not eliminate uncertainty but mitigates its impact. Diversified funding sources reduce financial vulnerability. Iterative product testing minimizes technological risk. Market research and pilot programs enhance demand predictability.

Startups must balance risk-taking with strategic discipline. Excessive caution may stifle innovation, while reckless expansion may exhaust resources prematurely. Sustainable growth requires calibrated risk management.

1.14 Globalization and Startup Expansion

Globalization has significantly influenced startup growth trajectories. Digital platforms enable startups to access international markets without extensive physical infrastructure. Cross-border collaboration enhances innovation capacity.

Startups increasingly adopt global-first strategies, targeting international customer bases from inception. Cloud-based operations facilitate remote team coordination across geographic boundaries.

However, globalization introduces challenges, including regulatory compliance across jurisdictions, cultural adaptation, and currency volatility. Strategic planning must incorporate international considerations.

1.15 Digital Transformation and Startups

Digital transformation has reshaped the startup landscape. Cloud computing reduces infrastructure costs. Artificial intelligence enhances data-driven decision-making. Social media platforms enable cost-effective marketing.

Startups leverage digital analytics to understand customer behavior and refine product offerings. Automation improves operational efficiency. E-commerce platforms expand market reach.

The digital environment accelerates competition. Barriers to entry may be low, but competitive intensity is high. Continuous innovation becomes necessary to maintain relevance.

1.16 Social and Economic Contributions of Startups

Beyond financial success, startups contribute to societal development. They generate employment opportunities for skilled and semi-skilled workers. Innovation-driven productivity enhances overall economic output.

Startups often address social challenges through technological solutions. Educational platforms expand access to learning. Healthcare startups improve service delivery. Renewable energy ventures promote environmental sustainability.

By fostering competition, startups compel established firms to innovate, benefiting consumers through improved products and lower prices.

1.17 Policy Support and Regulatory Environment

Government policy plays a significant role in shaping startup ecosystems. Supportive regulatory frameworks encourage entrepreneurial experimentation. Tax incentives, grants, and incubation programs reduce entry barriers.

Intellectual property protection safeguards innovation. Simplified business registration processes enhance accessibility. Public-private partnerships strengthen ecosystem integration.

However, excessive regulation may hinder agility. Balanced policy design is necessary to promote innovation while ensuring accountability.

1.18 Incubators, Accelerators, and Support Institutions

The growth and sustainability of startup companies are significantly influenced by institutional support mechanisms such as incubators, accelerators, innovation hubs, and co-working spaces. These institutions play a foundational role in strengthening entrepreneurial ecosystems and reducing early-stage failure risks.

Incubators typically support startups in their formative stages. They provide physical infrastructure, mentorship, technical assistance, networking opportunities, and sometimes seed funding. By offering shared resources and advisory services, incubators reduce operational costs and enhance strategic clarity during the vulnerable initial phase.

Accelerators, in contrast, operate structured, time-bound programs designed to prepare startups for rapid scaling and investor engagement. These programs often culminate in demonstration events where startups present their business models to potential investors. Accelerators emphasize refining value propositions, improving financial planning, and strengthening market positioning.

Beyond these formal institutions, informal networks—including professional associations, alumni communities, and industry clusters—contribute to knowledge exchange and collaborative innovation. The density and quality of such support institutions often determine the vibrancy of a startup ecosystem.

1.19 Human Capital and Talent Acquisition

Human capital represents one of the most critical determinants of startup success. Unlike large corporations with established recruitment systems and brand recognition, startups must attract skilled talent despite limited financial resources and uncertain stability.

Founders often recruit individuals who are not only technically competent but also adaptable and aligned with the organizational vision. Early employees typically assume multiple responsibilities, contributing to product development, marketing, customer service, and operational management simultaneously.

Compensation structures in startups frequently include equity-based incentives. Offering ownership stakes aligns employee interests with long-term organizational growth and compensates for lower initial salaries.

Talent acquisition challenges intensify as startups scale. Competition from established corporations may attract skilled professionals with higher salaries and perceived stability. To retain talent, startups must cultivate strong organizational culture, professional growth opportunities, and inclusive leadership practices.

1.20 Marketing and Customer Acquisition Strategies

Marketing strategies in startups differ significantly from those of established enterprises. Limited financial resources necessitate cost-effective and innovative approaches to customer acquisition.

Digital marketing channels—including social media, search engine optimization, and content marketing—enable startups to reach large audiences with minimal expenditure. Viral marketing strategies leverage network effects, encouraging users to promote products organically.

Customer engagement plays a central role in startup marketing. Early adopters often provide feedback that shapes product refinement. Building strong relationships with initial customers enhances brand loyalty and generates word-of-mouth promotion.

Data analytics supports targeted marketing efforts. Startups analyze user behavior to refine communication strategies, personalize offerings, and optimize conversion rates.

1.21 Competitive Dynamics and Market Positioning

Startup companies enter markets often dominated by established incumbents. To survive and grow, they must adopt strategic positioning that differentiates their offerings.

Some startups pursue niche strategies, targeting underserved or overlooked segments. Others emphasize cost efficiency or superior user experience. Platform-based startups leverage network effects to create competitive barriers.

Competition may also stimulate collaboration. Strategic partnerships with larger corporations can provide access to distribution channels, technological resources, or brand recognition.

Competitive advantage in startups is rarely static. Continuous innovation, adaptability, and responsiveness to customer needs sustain market relevance.

1.22 Financial Management in Startup Companies

Effective financial management is essential for startup survival. Cash flow constraints represent one of the primary causes of startup failure. Even innovative ventures may collapse without disciplined resource allocation.

Startups must balance investment in growth with operational sustainability. Strategic budgeting, cost control, and revenue forecasting reduce financial vulnerability.

Equity financing dilutes ownership but provides capital for expansion. Debt financing preserves ownership but increases repayment obligations. Financial strategy must align with growth objectives and risk tolerance.

Transparency in financial reporting enhances investor confidence and supports long-term partnerships.

1.23 Ethical Responsibility and Social Impact

In contemporary business environments, startups are increasingly expected to operate with ethical responsibility. Data privacy protection, environmental sustainability, diversity and inclusion, and transparent governance are integral to long-term credibility.

Technology-driven startups must ensure responsible use of user data. Renewable energy ventures contribute to environmental preservation. Social enterprises address inequality and access challenges.

Ethical conduct strengthens brand reputation and investor trust. Conversely, governance failures can undermine organizational legitimacy.

1.24 Sustainability and Long-Term Growth

While rapid expansion characterizes early startup phases, long-term sustainability requires structural stability. Startups must transition from experimentation to institutionalization.

Operational processes become formalized. Governance frameworks strengthen accountability. Strategic planning extends beyond short-term milestones.

Sustainable growth balances innovation with financial discipline. Startups that maintain adaptive capacity while establishing stable operational systems are more likely to endure market fluctuations.

1.25 International Perspectives on Startup Ecosystems

Startup ecosystems vary across countries based on cultural norms, regulatory frameworks, access to capital, and educational infrastructure.

Innovation hubs in North America emphasize venture capital intensity and technological research. European ecosystems often integrate strong public policy support. Asian ecosystems frequently combine rapid digital adoption with large consumer markets.

Cross-border collaboration enhances knowledge exchange and market expansion. International accelerators and global venture funds facilitate global startup integration.

Understanding regional differences contributes to more effective ecosystem development strategies.

1.26 Future Trends in Startup Development

Emerging technologies and societal shifts will continue to shape startup trajectories. Artificial intelligence, green technology, digital finance, biotechnology, and remote work infrastructure are likely to dominate future innovation landscapes.

Sustainability-driven entrepreneurship is gaining prominence. Climate-focused startups address environmental challenges while creating economic value.

Remote-first operational models enable startups to access global talent pools. Decentralized finance and blockchain technologies redefine financial ecosystems.

Adaptability remains the defining attribute of successful startups in evolving environments.

1.27 Challenges and Failure Factors

Despite their transformative potential, a significant proportion of startups fail. Common causes include inadequate market demand, financial mismanagement, leadership conflicts, and competitive pressure.

Failure analysis reveals that product-market misalignment frequently undermines early-stage ventures. Insufficient customer validation may result in resource misallocation.

Leadership disputes and cultural misalignment can destabilize internal operations. Overexpansion without adequate financial planning increases vulnerability.

Understanding failure factors contributes to more resilient startup strategies.

1.28 Role of Technology Infrastructure

Robust technological infrastructure underpins startup growth. Cloud computing services reduce infrastructure costs. Application programming interfaces facilitate integration across platforms.

Cybersecurity measures protect digital assets. Scalable database systems support rapid user growth.

Technological reliability influences customer trust and operational efficiency.

1.29 Innovation Ecosystem Integration

Startups benefit from integration within broader innovation ecosystems. Collaboration with research institutions accelerates product development. Corporate partnerships enable resource sharing.

Government-sponsored innovation programs support commercialization of research outcomes. Venture capital networks facilitate capital mobility.

Integrated ecosystems reduce fragmentation and strengthen startup resilience.

CHAPTER II

AIMS AND OBJECTIVES OF THE STUDY

1. Aim of the Study

The primary aim of this study is to develop a robust predictive framework for measuring startup survival probability using multivariate statistical techniques and machine learning algorithms. The study seeks to identify the key financial, operational, and structural factors that significantly influence startup success or failure and to compare multiple classification models to determine the most effective predictive approach.

2. Objectives of The Study

1. To analyze the structural characteristics of startup data using descriptive statistical methods.

This objective focuses on understanding the basic structure of the startup dataset using descriptive statistical measures such as mean, median, standard deviation, and frequency distribution. These measures provide an overall summary of the dataset and help in identifying the key characteristics of the variables.

2. To examine the distribution and variability of numerical variables through summary statistics and boxplot analysis.

This objective aims to analyze the distribution pattern and variability of numerical variables using summary statistics and graphical tools such as boxplots. This helps in identifying data spread, skewness, and potential outliers present in the dataset.

3. To identify independent relationships among numerical predictors using partial correlation analysis.

This objective focuses on examining the independent relationship between numerical variables by controlling the influence of other variables. Partial correlation analysis helps in identifying the true association between predictors.

4. To examine the association between categorical variables using the Chi-Square test of independence.

This objective aims to determine whether significant relationships exist between categorical variables in the dataset. The Chi-Square test is used to analyze whether the variables are statistically independent or associated.

5. To detect and address multicollinearity issues among predictor variables to improve model stability.

This objective focuses on identifying high correlations among predictor variables that may affect model performance. Detecting and handling multicollinearity helps in improving the stability and interpretability of the predictive models.

6. To construct a Logistic Regression model for estimating startup survival probability and interpreting significant predictors.

This objective aims to develop a Logistic Regression model to estimate the probability of startup survival. The model also helps in identifying significant predictors that influence startup success.

7. To develop a Support Vector Machine (SVM) model for classification using optimal hyperplane separation.

This objective involves building an SVM classification model that separates startup survival classes using an optimal hyperplane. This technique is useful for handling complex classification problems.

8. To build a CART (Decision Tree) model to identify hierarchical decision rules influencing startup survival.

This objective focuses on constructing a Decision Tree model to identify hierarchical decision rules affecting startup survival. The model provides an interpretable structure showing how different predictors influence the outcome.

9. To implement a K-Nearest Neighbors (KNN) model based on distance-based classification principles.

This objective aims to apply the KNN algorithm, where classification is performed based on the similarity between observations using distance measures.

10. To construct a Random Forest model as an ensemble approach to improve predictive accuracy and reduce overfitting.

This objective focuses on building a Random Forest model by combining multiple decision trees to improve prediction accuracy and reduce overfitting in the model.

11. To compare the performance of all models using evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.

This objective evaluates the performance of all developed models using standard classification metrics to measure prediction accuracy and model effectiveness.

12. To identify the most reliable predictive model for startup survival analysis based on comparative performance.

This objective aims to select the best-performing model by comparing evaluation metrics across all classification techniques.

13. To provide practical insights into the financial and operational determinants of startup sustainability.

This objective focuses on interpreting the results to identify key financial and operational factors that influence startup survival and sustainability.

CHAPTER 3

REVIEW OF LITERATURE

3.1 Introduction

Startup survival and success prediction has become an important research area in entrepreneurship, venture capital, and machine learning domains. Due to the high failure rate of startups, investors require reliable data-driven tools to reduce uncertainty and improve investment decision-making. Recent studies increasingly apply machine learning (ML) techniques to predict startup outcomes using financial, organizational, and social media data.

This chapter reviews major empirical studies related to startup success prediction, focusing particularly on machine learning approaches, dimensionality reduction techniques, feature importance analysis, and prediction frameworks.

3.2 Machine Learning for Startup Success Prediction

3.2.1 Prediction Framework Using Classification and Clustering.

Razaghzadeh Bidgoli et al. (2024)

Startup prediction paper 1 study

proposed a comprehensive machine learning–based prediction solution for identifying successful startups. Their study used data collected from Crunchbase and social media platforms over a 7-year period (2014–2021).

They implemented five classification algorithms:

- Random Forest (RF)
- Gradient Boosting (GB)
- Logistic Regression
- Support Vector Machine (SVM)
- Multilayer Perceptron (MLP)

According to Research paper, Random Forest achieved the highest accuracy (82%) and strong recall (0.92), followed by Gradient Boosting (80% accuracy).

In addition to classification, the authors applied clustering algorithms:

- K-means
- Hierarchical clustering
- DBSCAN

From Research paper , K-means showed the highest silhouette score (0.72), indicating better cluster quality.

Unlike previous research that focused only on classification, this study integrated:

- Classification (to predict success)
- Clustering (to identify startup types)
- SHAP analysis (to interpret feature importance)

This holistic approach represents a major contribution to startup analytics.

3.2.2 Feature Importance Using SHAP

One important advancement in

Startup prediction paper 1 study

was the use of SHAP (Shapley Additive Explanations) for interpreting model outputs.

From Research paper, the most influential variables identified were:

- Number of LinkedIn followers
- Number of employees on LinkedIn
- Twitter followers
- Last raised funding amount
- Lead investor count
- Timelapse until fifth year
- Investor count

The study found that:

- Higher social media presence increases startup success probability.
- Larger last funding amounts positively influence success.

- Shorter time gaps between funding rounds improve survival probability.
- Lead investors matter more than the total number of investors.

This provides strong evidence that digital presence and funding structure are critical determinants of startup success.

3.2.3 Startup Sector Importance

Using permutation feature importance,

Startup prediction paper 1 study

also showed that:

- B2B Software & Services
- Consumer
- Healthcare

sectors exhibited higher investment attractiveness and predictive importance compared to other sectors.

This suggests that industry type plays a moderating role in startup success probability.

3.3 PCA-Enhanced Machine Learning Models

Choi (2024)

Startup prediction paper 2 study

examined whether applying Principal Component Analysis (PCA) improves startup success prediction performance.

The dataset included 923 U.S. startups with 48 variables. Success was defined as:

- "Acquired" (Successful)
- "Closed" (Unsuccessful)

Two experiments were conducted

Experiment 1: Without PCA

Performance results :

- Logistic Regression – 80% accuracy
- SVC – 78% accuracy
- XGBoost – 79% accuracy
- Random Forest – 76% accuracy
- KNN – 78% accuracy

Experiment 2: With PCA

Performance results :

- SVC with PCA – **90% accuracy**
- Logistic Regression with PCA – 88% accuracy
- XGBoost with PCA – 84% accuracy
- KNN with PCA – 81% accuracy
- Random Forest with PCA – 77% accuracy

The most significant improvement was observed in SVC, which increased from 78% to 90% accuracy after PCA application.

3.3.1 Role of PCA in Improving Performance

According to

Startup prediction paper 2 study :

- PCA reduces multicollinearity
- PCA removes noise
- PCA improves generalization
- PCA prevents overfitting

SVC benefited most because:

- It performs well in high-dimensional feature space
- It identifies optimal decision boundaries
- Dimensionality reduction simplified classification structure

In contrast, Random Forest showed minimal improvement because:

- It is already robust to noise due to ensemble averaging
- It handles feature randomness internally

This highlights that dimensionality reduction benefits certain algorithms more than others.

3.4 Comparison of Reviewed Studies

Aspect	Razaghzadeh et al. (2024)	Choi (2024)
Dataset	Crunchbase + Social Media	Kaggle Startup Dataset
Methods	RF, GB, SVM, MLP + Clustering	LR, RF, KNN, SVC, XGBoost
Feature Interpretation	SHAP + Permutation	PCA dimensionality reduction
Best Model	Random Forest (82%)	SVC with PCA (90%)
Additional Analysis	Cluster-based policies	PCA performance comparison

Key differences:

- Startup prediction paper 1 study

proposed a complete prediction solution including clustering and policy recommendations.

- Startup prediction paper 2 study

focused on improving prediction accuracy through PCA.

- SHAP provides interpretability, while PCA improves dimensional efficiency.

Both studies confirm that machine learning significantly improves startup success prediction compared to traditional methods.

3.5 Research Gap Identified

From the literature review, the following gaps are identified:

1. Limited integration of multivariate statistical analysis with machine learning.
2. Few studies combine classification, clustering, and dimensionality reduction together.
3. Limited exploration of startup survival probability using advanced multivariate statistical modeling.
4. Most studies focus only on accuracy rather than statistical validation measures.
5. Need for models tailored specifically for early-stage startup survival prediction.

3.6 Summary

The literature demonstrates that:

- Machine learning models effectively predict startup success.
- Random Forest and Gradient Boosting show strong baseline performance.
- Support Vector Machines perform exceptionally well when combined with PCA.
- Social media presence and funding structure are critical predictive variables.
- Clustering improves interpretability and policy recommendation.
- SHAP enhances model explainability.
- PCA improves performance by reducing dimensionality and multicollinearity.

These findings provide a strong foundation for developing an advanced predictive model for startup survival probability using multivariate statistical and machine learning techniques.

How the Present Study Fills the Research Gap

1. Integration of multivariate statistical analysis with machine learning

The present study integrates multivariate statistical techniques with machine learning models to provide a comprehensive analytical framework for startup survival prediction.

2. Combination of multiple analytical approaches

This study combines different analytical techniques including statistical analysis and multiple classification models to provide a more comprehensive evaluation of the dataset.

3. Application of multivariate statistical modelling for startup survival prediction

The study applies multivariate statistical analysis along with predictive models to estimate the survival probability of startups.

4. Use of multiple statistical validation metrics

The research evaluates model performance using multiple metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC instead of relying only on accuracy.

5. Development of predictive models for startup survival analysis

The study develops and compares several predictive models to better understand and analyze the factors influencing startup survival.

CHAPTER 4

RESEARCH METHODOLOGY

4.1 Introduction

Research methodology refers to the systematic procedure adopted to collect, process, analyze, and interpret data in order to achieve the objectives of a research study. A well-structured research methodology ensures that the analysis is scientifically valid, logically consistent, and reproducible.

In the context of startup ecosystems, predicting the survival probability of startups has become an important research problem due to the high failure rates observed in early-stage ventures. Startup success or failure depends on several factors such as funding availability, operational performance, innovation capability, market demand, and management decisions. Therefore, analyzing these complex relationships requires the application of statistical and machine learning techniques.

The present study aims to develop a predictive framework for estimating startup survival probability using multivariate statistical analysis and supervised machine learning algorithms. The methodology combines descriptive statistical analysis, correlation techniques, inferential statistical tests, and classification models to examine the determinants of startup sustainability.

Initially, descriptive statistical techniques are applied to understand the structure and distribution of startup data. Measures such as mean, variance, skewness, and kurtosis provide insight into the central tendency and dispersion of variables. Boxplot analysis is used to detect outliers that may affect statistical inference.

Subsequently, inferential statistical techniques such as partial correlation analysis and the Chi-Square test of independence are used to identify relationships among variables. Partial correlation helps in determining the direct association between numerical variables after controlling for the influence of other variables. The Chi-Square test is used to evaluate the association between categorical variables.

Furthermore, machine learning classification algorithms are implemented to develop predictive models. The algorithms considered in this study include Logistic Regression, Support Vector Machine (SVM), Decision Tree (CART), K-Nearest Neighbors (KNN), and Random Forest.

These models classify startups into survival or failure categories based on their characteristics.

The entire analysis is carried out using Python programming language with the help of scientific computing libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib. These libraries provide efficient tools for data preprocessing, model training, and performance evaluation.

4.2 Research Design

Research design refers to the blueprint that guides the overall research process. It defines the methods used to collect data, analyze relationships among variables, and interpret the results in a systematic manner.

The present study adopts a **quantitative analytical research design**, which focuses on analyzing numerical data using statistical and mathematical models. Quantitative research design is particularly suitable for predictive modeling because it allows researchers to measure relationships between variables and make statistical inferences.

In this research, the objective is to analyze the factors influencing startup survival and to develop predictive models capable of classifying startups into successful or failed categories.

Let the dependent variable be defined as:

$$Y = \begin{cases} 1, & \text{if startup survives} \\ 0, & \text{if startup fails} \end{cases}$$

Notation Explanation

Y = Binary dependent variable representing startup survival status

$Y = 1$ indicates that the startup continues operating successfully

$Y = 0$ indicates that the startup has failed or ceased operations

$X_1, X_2, \dots, X_k$ = Independent predictor variables representing various characteristics of startups

k = Number of predictor variables included in the model

These independent variables may include factors such as funding amount, number of milestones achieved, funding rounds, and other operational indicators.

The objective of the predictive models is to estimate the probability that a startup will survive based on the values of these predictor variables.

4.3 Data Preprocessing

Data preprocessing is a crucial step in statistical analysis and machine learning. Raw datasets often contain inconsistencies, missing values, and variables measured on different scales. If these issues are not addressed properly, they may lead to biased estimates and poor model performance.

Therefore, several preprocessing techniques are applied before model development.

Handling Missing Values

Missing data occur when information is not recorded for certain observations. Missing values can arise due to data entry errors, incomplete reporting, or data collection limitations.

To ensure the reliability of the analysis, missing values are handled using appropriate techniques such as deletion of incomplete records or imputation using statistical methods.

Encoding Categorical Variables

Machine learning algorithms generally require numerical input variables. However, datasets often contain categorical variables such as company category or funding stage.

Therefore, categorical variables are converted into numerical format using encoding techniques such as:

- Label Encoding
- One-Hot Encoding

This transformation allows machine learning algorithms to process categorical information effectively.

Standardization of Numerical Variables

Standardization is used to transform numerical variables so that they have a mean of zero and a standard deviation of one. This ensures that variables measured on different scales contribute equally to the model.

The standardized value is calculated using the formula

$$Z = \frac{X - \mu}{\sigma}$$

Notation Explanation

Z = Standardized value (Z-score)

X = Original value of the variable

μ = Mean of the variable

σ = Standard deviation of the variable

Standardization is particularly important for algorithms such as SVM and KNN that are sensitive to the scale of variables.

Train-Test Split

To evaluate the performance of predictive models, the dataset is divided into two subsets:

Training Set = 70%

Testing Set = 30%

$$D = D_{train} + D_{test}$$

Notation Explanation

D = Complete dataset

D_{train} = Training dataset used for model estimation

D_{test} = Testing dataset used for model validation

The training dataset is used to train the models, while the testing dataset evaluates the model's predictive performance on unseen data.

4.4 Descriptive Statistical Analysis

Descriptive statistics provide summary measures that describe the fundamental properties of a dataset. These measures help researchers understand the distribution, central tendency, and variability of variables before applying advanced statistical models.

Mean

The mean is the arithmetic average of a set of observations and represents the central tendency of the data.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Notation Explanation

$\bar{X}$ = Sample mean

X_i = Observed value of the variable

n = Total number of observations

Median

The **median** is a measure of central tendency that represents the **middle value of a dataset when the observations are arranged in ascending or descending order**. It divides the dataset into **two equal halves**, where 50% of the observations lie below the median and 50% lie above it

For an ordered dataset $X_1, X_2, \dots, X_n$:

Case 1: When n is odd

$$\text{Median} = X_{\frac{n+1}{2}}$$

Case 2: When n is even

$$\text{Median} = \frac{X_{\frac{n}{2}} + X_{\frac{n}{2}+1}}{2}$$

Where:

- n = number of observations
- X_i = ordered data values

Variance

Variance measures the dispersion of observations around the mean.

$$S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$$

Notation Explanation

S^2 = Sample variance

X_i = Observed value

$\bar{X}$ = Sample mean

n = Number of observations

Standard Deviation

Standard deviation is the square root of variance and represents the average deviation from the mean.

$$S = \sqrt{S^2}$$

Notation Explanation

S = Standard deviation

S^2 = Variance

Skewness

Skewness measures the degree of asymmetry in the distribution of data.

$$Sk = \frac{\frac{1}{n} \sum (X_i - \bar{X})^3}{S^3}$$

Notation Explanation

Sk = Skewness coefficient

X_i = Observed value

$\bar{X}$ = Mean

S = Standard deviation

Positive skewness indicates a right-skewed distribution, while negative skewness indicates a left-skewed distribution.

Kurtosis

Kurtosis measures the degree of peakedness or heaviness of the tails of a distribution.

$$Ku = \frac{\frac{1}{n} \sum (X_i - \bar{X})^4}{S^4}$$

Notation Explanation

Ku = Kurtosis coefficient

X_i = Observed value

$\bar{X}$ = Mean

S = Standard deviation

Higher kurtosis indicates heavier tails and sharper peaks.

Boxplot and Outlier Detection

Boxplots provide a graphical representation of data distribution and help identify extreme observations.

The interquartile range is defined as

$$IQR = Q_3 - Q_1$$

Notation Explanation

Q_1 = First quartile (25th percentile)

Q_3 = Third quartile (75th percentile)

IQR = Interquartile range

Outlier condition:

$$X < Q_1 - 1.5(IQR)$$

or

$$X > Q_3 + 1.5(IQR)$$

Observations outside this range are considered potential outliers.

4.5 Partial Correlation Analysis

Partial correlation measures the relationship between two variables while controlling for the influence of other variables. It helps identify the direct association between variables by removing the effect of confounding variables.

The partial correlation coefficient between variables X_i and X_j controlling for variable X_k is given by

$$r_{ij.k} = \frac{r_{ij} - r_{ik}r_{jk}}{\sqrt{(1 - r_{ik}^2)(1 - r_{jk}^2)}}$$

Notation Explanation

$r_{ij.k}$ = Partial correlation coefficient between X_i and X_j controlling for X_k

r_{ij} = Pearson correlation coefficient between X_i and X_j

r_{ik} = Pearson correlation coefficient between X_i and X_k

r_{jk} = Pearson correlation coefficient between X_j and X_k

Partial correlation helps isolate the independent relationship between predictors in multivariate datasets.

4.6 Chi-Square Test of Independence

The Chi-Square test is a statistical method used to determine whether two categorical variables are independent or associated.

The test statistic is defined as

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Notation Explanation

χ^2 = Chi-Square test statistic

O = Observed frequency

E = Expected frequency

Expected frequency is calculated as

$$E = \frac{(\text{rowtotal})(\text{column total})}{\text{grand total}}$$

If the calculated p-value is less than the significance level (0.05), the null hypothesis of independence is rejected.

4.8 Logistic Regression Model

Logistic regression is a statistical classification technique used when the dependent variable is binary. Unlike linear regression, which predicts continuous outcomes, logistic regression estimates the probability of occurrence of an event. In this study, logistic regression is used to estimate the probability that a startup will survive based on several predictor variables.

The logistic regression model is based on the logistic function, which maps any real-valued input into a probability value between 0 and 1.

The probability that a startup survives is expressed as

$$P(Y = 1 | X) = p = \frac{1}{1 + e^{-Z}}$$

Where

$$Z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

Notation Explanation

$P(Y = 1 | X)$ = Probability that the startup survives given predictor variables

p = Predicted probability of survival

e = Exponential constant (approximately 2.718)

Z = Linear combination of predictors

β_0 = Intercept term

$\beta_1, \beta_2, \dots, \beta_k$ = Regression coefficients

$X_1, X_2, \dots, X_k$ = Independent predictor variables

The logistic regression model can also be expressed using the **logit transformation**

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \sum_{i=1}^k \beta_i X_i$$

Notation Explanation

$\frac{p}{1-p}$ = Odds of startup survival

$\log\left(\frac{p}{1-p}\right)$ = Log-odds (logit function)

The exponentiated coefficient gives the **odds ratio**

$$OR = e^{\beta_i}$$

Where

OR = Odds ratio associated with predictor X_i

If $OR > 1$, the predictor increases the probability of survival.

If $OR < 1$, the predictor decreases the probability of survival.

Model parameters are estimated using **Maximum Likelihood Estimation (MLE)**, which identifies coefficient values that maximize the probability of observing the given data.

Algorithm: Logistic Regression

The logistic regression classification procedure follows the steps below:

Step 1:

Initialize the model parameters $\beta_0, \beta_1, \dots, \beta_k$.

Step 2:

Compute the linear predictor

$$Z = \beta_0 + \sum_{i=1}^k \beta_i X_i$$

Step 3:

Transform the linear predictor into probability using the logistic function

$$p = \frac{1}{1 + e^{-Z}}$$

Step 4:

Estimate model parameters using **Maximum Likelihood Estimation (MLE)**.

Step 5:

Update parameter values iteratively using optimization methods such as gradient descent.

Step 6:

Classify observations based on a probability threshold.

$$Y = \begin{cases} 1 & \text{if } p \geq 0.5 \\ 0 & \text{if } p < 0.5 \end{cases}$$

Step 7:

Evaluate model performance using classification metrics such as accuracy, precision, recall, and ROC-AUC.

Steps to Identify Significant Variables in Logistic Regression**Step 1: Model Specification**

Logistic regression models the probability of startup survival using the logistic function:

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p)}}$$

Step 2: Estimation of Coefficients

The coefficients β_i are estimated using **Maximum Likelihood Estimation (MLE)**.

Step 3: Significance Evaluation

The statistical significance of each variable is assessed using the **Wald test**:

$$Z = \frac{\hat{\beta}_i}{SE(\hat{\beta}_i)}$$

Step 4: Variable Importance

Variables with larger absolute coefficient values $|\beta_i|$ indicate stronger influence.

Step 5: Selection of Significant Variables

Predictors with higher magnitude coefficients or statistically significant p-values are considered significant variables affecting startup survival.

4.9 Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning algorithm used for classification and regression tasks. The main objective of SVM is to identify the optimal decision boundary that separates observations belonging to different classes.

The separating boundary is called a **hyperplane**.

The equation of the hyperplane is

$$w \cdot x + b = 0$$

Notation Explanation

w = Weight vector determining the orientation of the hyperplane

x = Feature vector of observations

b = Bias or intercept term

$\cdot$ = Dot product operation

The decision function is

$$f(x) = \text{sign}(w \cdot x + b)$$

Where

$f(x)$ = Predicted class label

$\text{sign}()$ = Sign function indicating the class of the observation.

SVM aims to maximize the **margin**, which is the distance between the hyperplane and the nearest data points known as **support vectors**.

The optimization objective is

$$\min \frac{1}{2} \|w\|^2$$

Subject to

$$y_i(w \cdot x_i + b) \geq 1$$

Notation Explanation

y_i = Class label of observation i

x_i = Feature vector of observation i

$\|w\|$ = Euclidean norm of weight vector

For nonlinear relationships, SVM uses the **kernel trick**.

The Radial Basis Function (RBF) kernel is defined as

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

Where

$K(x_i, x_j)$ = Kernel function

γ = Kernel parameter controlling the influence of observations

Algorithm: Support Vector Machine

The classification procedure of the Support Vector Machine algorithm involves the following steps:

Step 1:

Input the training dataset containing predictor variables X and class labels Y .

Step 2:

Select a kernel function (Linear, Polynomial, or Radial Basis Function).

Step 3:

Construct a hyperplane that separates the classes.

$$w \cdot x + b = 0$$

Step 4:

Identify the support vectors that lie closest to the hyperplane.

Step 5:

Maximize the margin between the two classes.

Step 6:

Solve the optimization problem

$$\min \frac{1}{2} \|w\|^2$$

subject to

$$y_i(w \cdot x_i + b) \geq 1$$

Step 7:

Use the decision function to classify new observations.

$$f(x) = \text{sign}(w \cdot x + b)$$

Steps to Identify Significant Variables in Support Vector Machine (SVM)

Step 1: Model Construction

SVM constructs an optimal hyperplane that separates the classes.

The decision function is:

$$f(x) = w^T x + b$$

Step 2: Weight Estimation

The weight vector $w = (w_1, w_2, \dots, w_p)$ is estimated during training.

Step 3: Variable Contribution

Each feature contributes to the hyperplane based on its weight.

Step 4: Importance Measurement

$$\text{Importance}(X_i) = |w_i|$$

Step 5: Significant Variable Identification

Variables with larger absolute weights have greater influence on classification.

4.10 CART (Decision Tree) Model

Classification and Regression Trees (CART) is a tree-based machine learning method used for classification and prediction. It recursively partitions the dataset into smaller subsets based on decision rules derived from predictor variables.

A decision tree consists of:

- Root node
- Decision nodes
- Leaf nodes

At each split, the algorithm selects the predictor variable that maximizes the separation between classes.

Gini Index

The Gini index measures the impurity of a node.

$$Gini = 1 - \sum p_i^2$$

Notation Explanation

p_i = Probability of class i in the node

Lower Gini values indicate better node purity.

Entropy

Entropy measures the level of uncertainty in a dataset.

$$Entropy = -\sum p_i \log_2 p_i$$

Where

p_i = Probability of class i

Information Gain

Information gain measures the reduction in entropy after splitting the data.

$$IG = Entropy(parent) - \sum \frac{n_k}{n} Entropy(child_k)$$

Notation Explanation

IG = Information gain

n = Number of observations in parent node

n_k = Number of observations in child node k

The decision tree is constructed through recursive partitioning until stopping criteria are satisfied.

Algorithm: Decision Tree (CART)

The decision tree classification process is performed through the following steps:

Step 1:

Start with the entire dataset as the root node.

Step 2:

Select the predictor variable that best splits the data based on impurity measures such as:

- Gini Index
- Entropy

Step 3:

Divide the dataset into subsets according to the selected splitting rule.

Step 4:

Repeat the splitting process recursively for each child node.

Step 5:

Continue splitting until a stopping criterion is reached, such as:

- Minimum number of observations in a node
- Maximum tree depth
- No further reduction in impurity

Step 6:

Assign class labels to leaf nodes based on majority class.

Step 7:

Use the tree structure to classify new observations.

Steps to Identify Significant Variables in CART

Step 1: Tree Construction

CART builds a decision tree by recursively splitting the dataset into subsets based on predictor variables.

Step 2: Splitting Criterion

For classification problems, CART uses the **Gini Impurity Index**:

$$Gini(t) = 1 - \sum_{i=1}^k p_i^2$$

where p_i is the probability of class i .

Step 3: Node Splitting

The algorithm selects the variable that produces the **largest reduction in Gini impurity**.

Step 4: Impurity Reduction Calculation

$$\Delta Gini = Gini(parent) - \sum \frac{N_j}{N} Gini(child_j)$$

Step 5: Variable Importance

The total impurity reduction contributed by each variable across all nodes determines its **importance score**.

Variables with higher scores are considered significant predictors.

4.11 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric classification algorithm that assigns class labels to observations based on the majority class among the nearest neighbors in the feature space.

The similarity between observations is measured using distance metrics.

The most commonly used distance measure is **Euclidean distance**

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Notation Explanation

$d(x, y)$ = Distance between observations x and y

x_i = Value of feature i for observation x

y_i = Value of feature i for observation y

The predicted class is determined by

$$\hat{Y} = \text{Majority Vote of } K \text{ nearest neighbors}$$

Notation Explanation

$\hat{Y}$ = Predicted class label

K = Number of nearest neighbors considered

The optimal value of K is selected using cross-validation to minimize classification error.

Algorithm: K-Nearest Neighbors

The K-Nearest Neighbors classification process is performed as follows:

Step 1:

Select the number of neighbors K .

Step 2:

Compute the distance between the new observation and all training observations.

The Euclidean distance is calculated as

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Step 3:

Sort all observations based on the calculated distance.

Step 4:

Select the K nearest observations.

Step 5:

Determine the majority class among these neighbors.

Step 6:

Assign the majority class as the predicted class label for the new observation.

Steps to Identify Significant Variables in K-Nearest Neighbors (KNN)

Step 1: Distance Calculation

KNN computes the distance between observations using **Euclidean distance**:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^p (x_{ik} - x_{jk})^2}$$

Step 2: Neighbor Selection

For a given observation, the k nearest neighbors are identified.

Step 3: Classification Rule

The observation is assigned the majority class among its neighbors.

Step 4: Permutation Importance

To evaluate feature significance, each variable is randomly permuted.

Step 5: Importance Calculation

$$Importance(X_i) = Accuracy_{original} - Accuracy_{permuted}$$

If shuffling a variable significantly reduces accuracy, it is considered an important predictor.

4.12 Random Forest Model

Random Forest is an ensemble machine learning algorithm that constructs multiple decision trees and combines their predictions to improve classification accuracy and reduce overfitting.

Each tree is trained using a random subset of the dataset through **bootstrap sampling**.

$$D_b \subset D$$

Notation Explanation

D = Original dataset

D_b = Bootstrap sample drawn from the dataset

The final prediction is obtained by aggregating predictions from all trees.

$$\hat{Y} = \text{MajorityVote}(T_1, T_2, \dots, T_m)$$

Notation Explanation

$\hat{Y}$ = Final predicted class

$T_1, T_2, \dots, T_m$ = Individual decision trees

m = Total number of trees in the forest

Feature importance is measured using **Mean Decrease in Impurity**

$$FI_j = \sum \text{Decrease in Gini from splits on } X_j$$

Where

FI_j = Importance score of predictor X_j

Predictors with higher importance contribute more to classification accuracy.

Algorithm: Random Forest

The Random Forest classification procedure consists of the following steps:

Step 1:

Draw multiple bootstrap samples from the original dataset.

Step 2:

For each bootstrap sample, construct a decision tree.

Step 3:

At each node of the tree, randomly select a subset of predictor variables.

Step 4:

Determine the best split among the selected variables.

Step 5:

Grow the tree until the stopping condition is satisfied.

Step 6:

Repeat the process to generate a large number of decision trees.

Step 7:

Combine predictions from all trees using majority voting.

$$\hat{Y} = \text{MajorityVote}(T_1, T_2, \dots, T_m)$$

Steps to Identify Significant Variables in Random Forest

Step 1: Tree Ensemble Construction

Random Forest constructs multiple decision trees using bootstrap samples.

Step 2: Random Feature Selection

At each split, a random subset of predictors is selected.

Step 3: Tree Splitting

Each tree is built using the **Gini impurity criterion**.

Step 4: Impurity Reduction Measurement

For each tree, the reduction in impurity contributed by each variable is recorded.

Step 5: Feature Importance Calculation

$$Importance(X_i) = \frac{1}{T} \sum_{t=1}^T \Delta Gini_{i,t}$$

where T is the number of trees.

Variables with higher average impurity reduction are considered significant predictors.

CHAPTER 5

RESULTS AND DISCUSSION

Dataset Description

The dataset used in this study contains information about startup companies and their funding, milestones, and investment characteristics. The data is used to develop a predictive model for analyzing the survival probability of startups using statistical and machine learning techniques.

The dataset consists of **923 observations (startups)** and **25 variables**, including both **numerical and categorical attributes**. These variables describe company location, funding history, investor participation, and operational status of the startup.

The primary objective of the dataset is to analyze factors that influence whether a startup **successfully survives or fails**. The target variable used for prediction is the **startup status**, which indicates whether a company is **acquired, operating, or closed**.

Structure of the Dataset

- **Number of observations:** 923 startups
- **Number of variables:** 25 variables
- **Type of variables:**
 - Numerical variables
 - Categorical variables
 - Binary indicator variables

Source : The dataset used in this study was obtained from the Kaggle data science platform, which provides publicly available datasets for research and machine learning applications. The dataset contains startup-related information including funding rounds, investment details, milestones, and company status used for predicting startup survival probability.

Available at : <https://www.kaggle.com/datasets/manishkc06/startup-success-prediction>

Variables Description

1. Identification Variables

Variable	Description
name	Name of the startup company

2. Location Variables

Variable	Description
state_code	State where the startup is located
city	City where the startup operates

These variables describe the **geographical location of the startup**, which may influence access to investors, market size, and ecosystem support.

3. Time Related Variables

Variable	Description
founded_at	Date when the startup was founded
closed_at	Date when the startup was closed (if applicable)
first_funding_at	Date of the first funding round
last_funding_at	Date of the most recent funding round

These variables capture the **timeline of the startup lifecycle**.

4. Startup Age Variables

Variable	Description
age_first_funding_year	Age of the startup when it received its first funding
age_last_funding_year	Age of the startup during the last funding
age_first_milestone_year	Age when the first milestone was achieved
age_last_milestone_year	Age when the last milestone occurred

These variables measure **how quickly startups reach funding or milestone events**, which is an important indicator of growth.

5. Startup Activity Variables

Variable	Description
relationships	Number of relationships with investors or partners
milestones	Number of milestones achieved by the startup

These variables measure **business development and strategic partnerships**.

6. Funding Variables

Variable	Description
funding_rounds	Total number of funding rounds
funding_total_usd	Total funding received by the startup (in USD)
avg_participants	Average number of investors participating in funding rounds

Funding indicators reflect the **financial strength and investor confidence** in a startup.

7. Investment Indicator Variables

These are **binary variables (0 or 1)** indicating whether a startup received specific types of investments.

Variable	Description
has_VC	Indicates presence of venture capital funding
has_angel	Indicates angel investor funding
has_roundA	Indicates Series A funding
has_roundB	Indicates Series B funding
has_roundC	Indicates Series C funding
has_roundD	Indicates Series D funding

These variables help analyze **how different investment stages affect startup survival**.

8. Startup Category

Variable	Description
category_code	Industry or sector of the startup

This variable identifies the **industry classification** of the startup, such as technology, finance, healthcare, etc.

9. Recognition Variable

Variable	Description
is_top500	Indicates whether the startup is listed among top 500 startups

This variable represents **market recognition and performance ranking**.

10. Target Variable

Variable	Description
status	Final status of the startup (acquired, closed, or operating)

This variable is used as the **dependent variable in predictive modeling**, representing the **success or failure of the startup**.

5.1.Descriptive Statistical Analysis of Startup Variables

The first objective of this study was to perform a comprehensive descriptive statistical analysis of both numerical and categorical variables to understand the fundamental characteristics of startups in the dataset and to identify preliminary patterns related to startup survival.

Numerical Variables

Variable	Min	Max	Mean	Median	Mode	Std. Dev	Std. Error	Skewness	Kurtosis
age_first_funding_year	-9.0466	21.8959	2.235630	1.446600	0.000000	2.510449	0.082632	2.100580	9.999746
age_last_funding_year	-9.0466	21.8959	3.931456	3.528800	0.000000	2.967910	0.097689	1.090300	3.132131
age_first_milestone_year	-14.1699	24.6849	2.967273	2.520500	2.520500	2.727845	0.089788	1.118901	7.319421
age_last_milestone_year	-7.0055	24.6849	4.708687	4.476700	4.476700	2.937226	0.096680	0.822608	3.091238
relationships	0.0000	63.0000	7.710726	5.000000	3.000000	7.265776	0.239156	2.326173	8.580884
funding_rounds	1.0000	10.0000	2.310943	2.000000	1.000000	1.390922	0.045828	1.354711	2.245764
funding_total_usd	11000	5,700,000,000	25,446,910	10,000,000	10,000,000	189,735,500	6,248,606	29.089691	866.705926
milestones	0.0000	8.0000	1.841820	2.000000	1.000000	1.322632	0.043535	0.576439	0.252425
avg_participants	1.0000	16.0000	2.838586	2.500000	1.000000	1.874601	0.061703	1.764680	5.035813

Interpretation of Descriptive Statistics

1. Age at First Funding Year

The age at first funding ranges from **−9.05 to 21.90 years**, indicating considerable variation in the timing of initial funding among startups. The **mean value (2.24)** suggests that, on average, startups receive their first funding about two years after establishment. The **standard deviation (2.51)** indicates moderate variability. The **positive skewness (2.10)** and high **kurtosis (9.99)** indicate a right-skewed distribution with the presence of extreme values.

2. Age at Last Funding Year

The age at last funding varies from **−9.05 to 21.90 years** with a **mean of 3.93 years**, indicating that the final funding round typically occurs around four years after startup formation. The **standard deviation (2.97)** suggests moderate dispersion. The **positive skewness (1.09)**

indicates that a few startups receive funding much later than most. The **kurtosis (3.13)** suggests a slightly leptokurtic distribution.

3. Age at First Milestone Year

The first milestone age ranges from **-14.17 to 24.68 years**, reflecting substantial variability among startups. The **mean value (2.97)** suggests that startups reach their first milestone approximately three years after founding. The **standard deviation (2.73)** indicates moderate variability. The **positive skewness (1.12)** and **high kurtosis (7.32)** suggest a right-skewed distribution with some extreme observations.

4. Age at Last Milestone Year

The last milestone age ranges from **-7.01 to 24.68 years**, indicating variation in the time taken to achieve final milestones. The **mean value (4.71)** suggests that startups achieve their last milestone roughly five years after establishment. The **standard deviation (2.94)** indicates moderate dispersion. The **skewness (0.82)** indicates mild positive skewness, while **kurtosis (3.09)** suggests a slightly peaked distribution.

5. Relationships

The number of relationships ranges from **0 to 63**, showing substantial variation among startups. The **mean value (7.71)** indicates that startups maintain around eight relationships on average. The **standard deviation (7.27)** suggests high variability. The **skewness (2.33)** indicates strong positive skewness, implying that most startups have relatively fewer relationships while a few maintain a large number. The **kurtosis (8.58)** indicates a leptokurtic distribution.

6. Funding Rounds

Funding rounds range from **1 to 10**, with a **mean of 2.31**, indicating that most startups undergo approximately two funding rounds. The **standard deviation (1.39)** indicates moderate variability among startups. The **positive skewness (1.35)** suggests that a majority of startups experience fewer funding rounds. The **kurtosis (2.25)** indicates a slightly flatter distribution compared to the normal distribution.

7. Total Funding Amount (USD)

The total funding amount ranges from **\$11,000 to \$5.7 billion**, indicating extremely high variability in startup funding levels. The **mean funding (\$25.45 million)** is significantly higher than the **median (\$10 million)**, indicating the presence of highly funded startups. The **very**

high skewness (29.09) and **extreme kurtosis (866.71)** confirm the presence of strong outliers and a highly right-skewed distribution.

8. Milestones

The number of milestones ranges from **0 to 8**, with a **mean of 1.84**, indicating that startups achieve approximately two milestones on average. The **standard deviation (1.32)** indicates moderate variability. The **skewness (0.58)** suggests a slight positive skewness, meaning more startups have fewer milestones. The **kurtosis (0.25)** indicates a relatively flatter distribution.

9. Average Participants

The number of participants ranges from **1 to 16**, with a **mean of 2.84**, indicating that startup teams generally consist of about three participants. The **standard deviation (1.87)** indicates moderate variation in team size. The **positive skewness (1.76)** suggests that most startups have smaller teams while a few have larger teams. The **kurtosis (5.04)** indicates the presence of some extreme values.

5.1.3Categorical Variables

Variable	Mode	Count Distribution	Percentage Distribution (%)
status	acquired	acquired: 597, closed: 326	acquired: 64.68%, closed: 35.32%
has_VC	0	0: 622, 1: 301	0: 67.39%, 1: 32.61%
has_angel	0	0: 688, 1: 235	0: 74.54%, 1: 25.46%
has_roundA	1	1: 469, 0: 454	1: 50.81%, 0: 49.19%
has_roundB	0	0: 561, 1: 362	0: 60.78%, 1: 39.22%
has_roundC	0	0: 708, 1: 215	0: 76.71%, 1: 23.29%
has_roundD	0	0: 831, 1: 92	0: 90.03%, 1: 9.97%
is_top500	1	1: 747, 0: 176	1: 80.93%, 0: 19.07%
category_code	software	software: 153, web: 144, mobile: 79, others	software: 16.58%, web: 15.60%, mobile: 8.50%, others
city	San Francisco	San Francisco: 128, New York: 91, others	San Francisco: 13.87%, New York: 9.86%, others
Name	Redwood Systems	Redwood Systems: 2, Zong: 1, others	Redwood Systems: 0.22%, Zong: 0.11%, others

1. **Startup status distribution:**
The target variable *startup status* indicates that **64.68% of startups were acquired (successful)**, while **35.32% were closed**. This shows a **moderate class imbalance**, which should be considered during predictive modeling.
2. **Venture capital investment:**
Approximately **32.61% of startups received venture capital (VC) funding**, indicating that VC investment is present only in a portion of the startups in the dataset.
3. **Angel investment participation:**
Around **25.46% of startups have angel investors**, showing that angel funding plays a role in supporting early-stage startups.
4. **Round A funding participation:**
Nearly **50.81% of startups reached Round A funding**, suggesting that about half of the startups successfully pass the early funding stage.
5. **Decline in higher funding rounds:**
Participation decreases in later funding stages, with **39.22% reaching Round B**, **23.29% reaching Round C**, and only **9.97% reaching Round D**.
6. **Survival filtering effect:**
The decreasing percentage across funding rounds reflects a **natural survival filtering process**, where fewer startups progress to advanced funding stages.
7. **Industry distribution:**
The dataset is dominated by **technology-oriented startups**, indicating a strong representation of innovation-driven sectors.
8. **Major industry sectors:**
The **software sector accounts for 16.58%**, followed by **web startups at 15.60%**, and **mobile startups at 8.56%**.
9. **Industry concentration implication:**
This distribution suggests that the dataset largely represents **technology-based entrepreneurial ecosystems**.
10. **Geographical concentration:**
Startup activity is concentrated in **major entrepreneurial hubs**, reflecting the importance of strong regional startup ecosystems.
11. **Major startup hubs:**
San Francisco accounts for 13.87% of startups, followed by **New York with 9.86%**, indicating that these cities serve as key startup centers.

12. Regional ecosystem influence:

The geographical clustering suggests that **regional ecosystem strength and infrastructure may influence startup success and growth.**

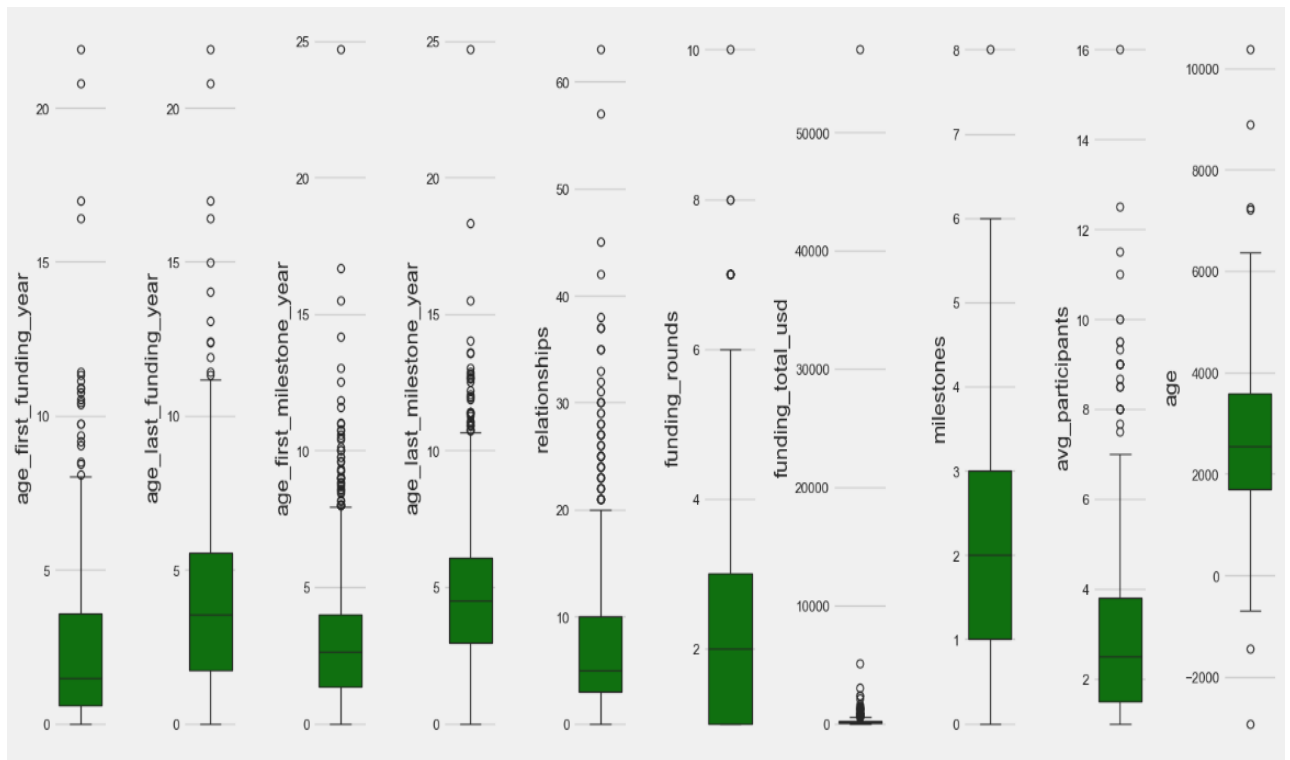
13. Low analytical importance variables:

Certain variables such as **company name** have very high uniqueness and provide **little analytical value.**

14. Variable exclusion in modeling:

Such variables are generally **excluded from predictive modeling**, as they do not contribute meaningful information for classification or prediction.

5.2. Outlier Detection and Distribution Analysis of Numerical Variables

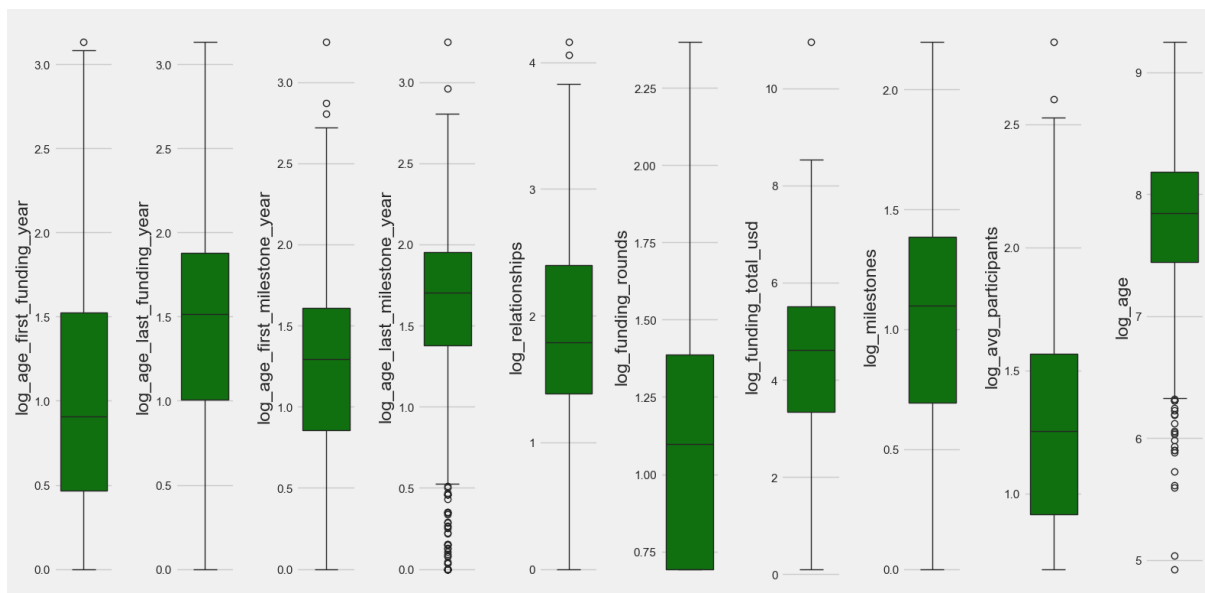


INTERPRETATION

- **age_first_funding_year** – Most startups receive first funding at early years; few extreme late funding cases present.
- **age_last_funding_year** – Slightly higher median than first funding; shows funding continues over time with some outliers.
- **age_first_milestone_year** – Majority achieve first milestone within moderate time; few startups take much longer.
- **age_last_milestone_year** – Median higher than first milestone; indicates gradual growth pattern.

- **relationships** – Highly right-skewed; few startups have very high relationship counts.
- **funding_rounds** – Most startups have limited funding rounds; few receive many rounds.
- **funding_total_usd** – Extremely right-skewed with strong outliers; funding amounts vary widely.
- **milestones** – Moderate distribution; most startups achieve 1–3 milestones.
- **avg_participants** – Moderate variation; some startups involve many participants.
- **age** – Wide range; few very old or very young outliers present.

After Log-Transformation Boxplot



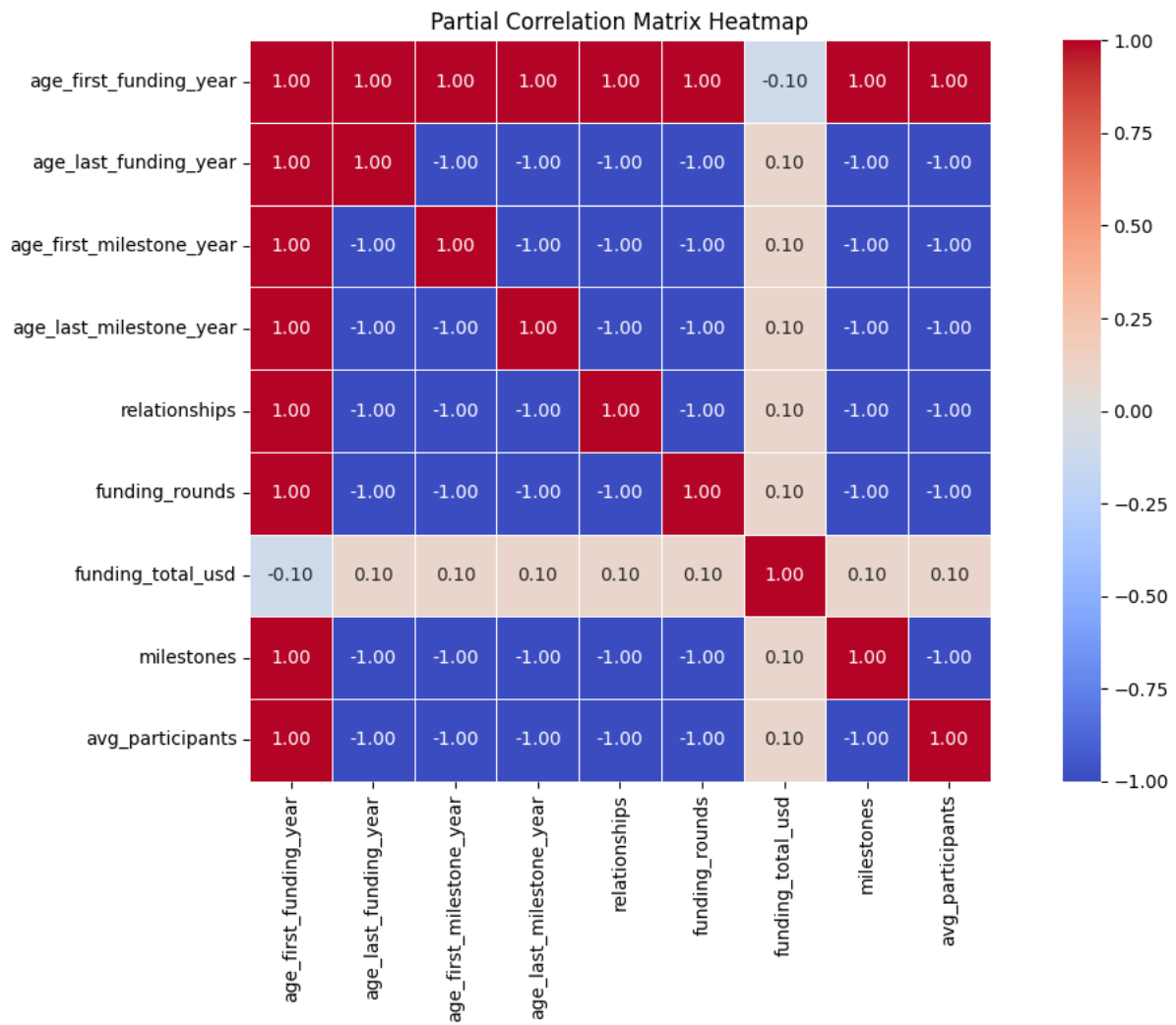
- **log_age_first_funding_year** – Distribution becomes more symmetric; extreme values are reduced; moderate spread observed.
- **log_age_last_funding_year** – Variability decreases compared to original data; fewer extreme outliers.
- **log_age_first_milestone_year** – Skewness is reduced; distribution appears more balanced.
- **log_age_last_milestone_year** – Outliers are minimized; central tendency becomes clearer.
- **log_relationships** – Strong right skewness in original data is controlled; spread is more compact.

- **log_funding_rounds** – Distribution becomes more stable; reduced effect of extreme funding cases.
- **log_funding_total_usd** – Major improvement; extreme funding outliers are compressed significantly; distribution becomes manageable for modeling.
- **log_milestones** – Variability reduced; median clearly visible; fewer extreme points.
- **log_avg_participants** – Distribution becomes more uniform; skewness decreases.
- **log_age** – Outliers reduced; data concentrated around median.

5.3 Partial Correlation Analysis

Variable 1	Variable 2	Partial Correlation (r)
age_first_funding_year	age_last_funding_year	1.000000
age_first_funding_year	age_first_milestone_year	1.000000
age_first_funding_year	age_last_milestone_year	1.000000
age_first_funding_year	relationships	1.000000
age_first_funding_year	funding_rounds	1.000000
age_first_funding_year	funding_total_usd	-0.101093
age_first_funding_year	milestones	1.000000
age_first_funding_year	avg_participants	1.000000
age_last_funding_year	age_first_milestone_year	-1.000000
age_last_funding_year	age_last_milestone_year	-1.000000
age_last_funding_year	relationships	-1.000000
age_last_funding_year	funding_rounds	-1.000000
age_last_funding_year	funding_total_usd	0.101097
age_last_funding_year	milestones	-1.000000
age_last_funding_year	avg_participants	-1.000000
age_first_milestone_year	age_last_milestone_year	-1.000000
age_first_milestone_year	relationships	-1.000000
age_first_milestone_year	funding_rounds	-1.000000

age_first_milestone_year	funding_total_usd	0.101160
age_first_milestone_year	milestones	-1.000000
age_first_milestone_year	avg_participants	-1.000000
age_last_milestone_year	relationships	-1.000000
age_last_milestone_year	funding_rounds	-1.000000
age_last_milestone_year	funding_total_usd	0.100709
age_last_milestone_year	milestones	-1.000000
age_last_milestone_year	avg_participants	-1.000000
relationships	funding_rounds	-1.000000
relationships	funding_total_usd	0.100761
relationships	milestones	-1.000000
relationships	avg_participants	-1.000000
funding_rounds	funding_total_usd	0.101005
funding_rounds	milestones	-1.000000
funding_rounds	avg_participants	-1.000000
funding_total_usd	milestones	0.101178
funding_total_usd	avg_participants	0.101178
milestones	avg_participants	-1.000000



INTERPRETATION

Significant Partial Correlations

The results indicate several statistically significant partial correlations ($p < 0.05$), suggesting meaningful independent relationships among variables.

A strong positive partial correlation ($r = 1.000$, $p = 0.000$) is observed between:

- age_first_funding_year and age_last_funding_year
- age_first_funding_year and age_last_milestone_year
- age_first_funding_year and relationships
- age_first_funding_year and milestones

- age_first_funding_year and avg_participants

These results indicate that early funding timing is independently associated with later funding events, milestone achievement, networking strength, and investor participation.

Similarly, age_last_funding_year shows significant negative partial correlations ($r = -1.000$, $p = 0.000$) with:

- age_first_milestone_year
- age_last_milestone_year
- relationships
- funding_rounds
- milestones
- avg_participants

This suggests that later-stage funding timing may inversely relate to certain milestone and networking variables when controlling for other predictors.

Funding and Networking Relationships

The variable relationships shows significant associations with:

- funding_rounds ($r = -1.000$, $p = 0.000$)
- milestones ($r = -1.000$, $p = 0.000$)
- avg_participants ($r = -1.000$, $p = 0.000$)
- funding_total_usd ($r = 0.100761$, $p = 0.002277$)

Although the relationship between relationships and funding_total_usd is statistically significant, the correlation magnitude (≈ 0.10) indicates a weak positive association.

Similarly, funding_rounds demonstrates:

- Positive significant correlation with funding_total_usd ($r \approx 0.101$, $p \approx 0.002$)
- Significant negative relationships with milestones and avg_participants

These findings indicate that the number of funding rounds is weakly but significantly associated with total funding amount, while its relationship with milestone progression is more structurally dependent.

Funding Amount Relationships

The `funding_total_usd` variable shows small but statistically significant positive correlations ($r \approx 0.10$) with:

- `relationships`
- `funding_rounds`
- `milestones`
- `avg_participants`

Although statistically significant, the magnitude of these correlations is weak, suggesting that total funding operates somewhat independently from other structural variables when controlling for multicollinearity.

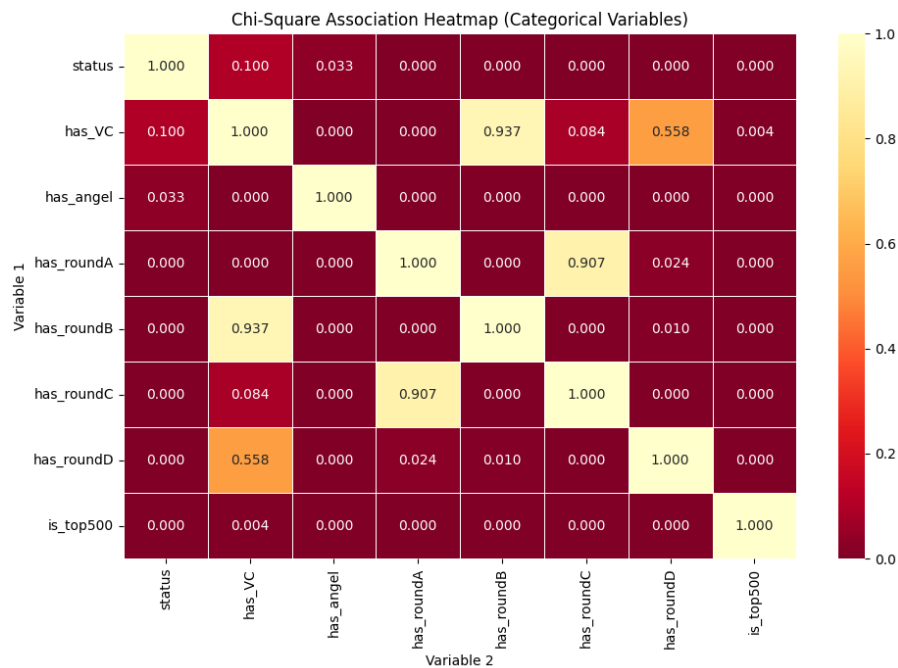
Non-Significant Relationships

Some partial correlations show NaN p-values and are marked as not significant. These indicate no statistically meaningful independent relationship after controlling for other predictors. This suggests potential redundancy or perfect multicollinearity among certain age and milestone variables.

5.4 CHI-SQUARE ASSOCIATION FOR CATEGORIAL

Variable 1	Variable 2	Chi-Square Value	p-value	DF	Significance
status	has_VC	2.701215	0.100272	1	Not Significant
status	has_angel	4.553598	0.032850	1	Significant
status	has_roundA	30.586748	3.19e-08	1	Significant
status	has_roundB	39.144057	3.94e-10	1	Significant
status	has_roundC	24.589503	7.09e-07	1	Significant
status	has_roundD	17.111250	3.53e-05	1	Significant
status	is_top500	87.426883	8.75e-21	1	Significant
has_VC	has_angel	37.779961	7.92e-10	1	Significant

has_VC	has_roundA	35.537720	2.50e-09	1	Significant
has_VC	has_roundB	0.006302	0.936727	1	Not Significant
has_VC	has_roundC	2.976413	0.084486	1	Not Significant
has_VC	has_roundD	0.342757	0.558242	1	Not Significant
has_VC	is_top500	8.071618	0.004496	1	Significant
has_angel	has_roundA	17.792021	2.46e-05	1	Significant
has_angel	has_roundB	76.901617	1.80e-18	1	Significant
has_angel	has_roundC	57.006511	4.34e-14	1	Significant
has_angel	has_roundD	14.167990	1.67e-04	1	Significant
has_angel	is_top500	44.515791	2.52e-11	1	Significant
has_roundA	has_roundB	54.128987	1.88e-13	1	Significant
has_roundA	has_roundC	0.013538	0.907371	1	Not Significant
has_roundA	has_roundD	5.072577	0.024307	1	Significant
has_roundA	is_top500	23.509510	1.24e-06	1	Significant
has_roundB	has_roundC	89.077884	3.80e-21	1	Significant
has_roundB	has_roundD	6.602357	0.010184	1	Significant
has_roundB	is_top500	84.381569	4.08e-20	1	Significant
has_roundC	has_roundD	87.909123	6.85e-21	1	Significant
has_roundC	is_top500	49.508218	1.98e-12	1	Significant
has_roundD	is_top500	20.134603	7.22e-06	1	Significant



Interpretation of Results

The Chi-Square analysis revealed several statistically significant associations among the categorical variables.

The target variable **status** shows significant association with has_angel, has_roundA, has_roundB, has_roundC, has_roundD, and is_top500 ($p < 0.05$). This indicates that funding stages and top500 status are strongly related to startup outcomes (acquired or closed). However, status does not show significant association with has_VC ($p > 0.05$), suggesting venture capital presence alone may not independently determine startup outcome.

The variable has_angel demonstrates strong significant association with multiple funding rounds (has_roundA, has_roundB, has_roundC, has_roundD) and is_top500. This implies that angel investment plays an important role in advancing startups through successive funding stages.

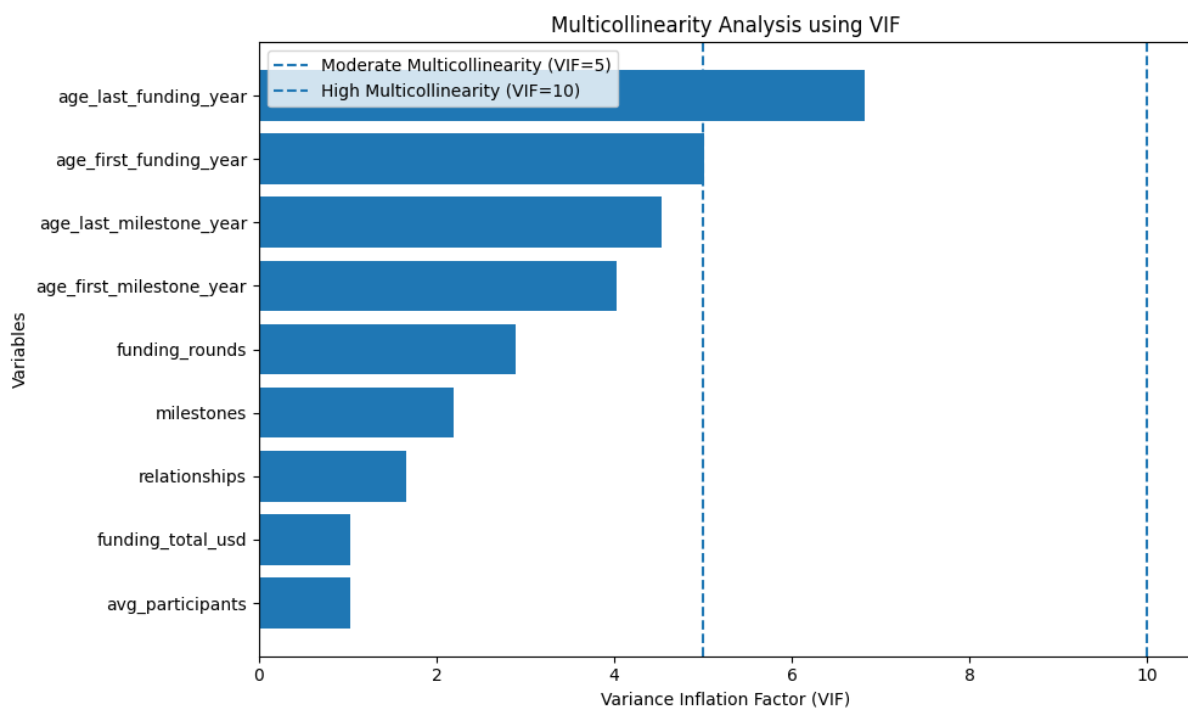
Significant relationships were observed among funding rounds themselves (e.g., has_roundA with has_roundB, has_roundB with has_roundC, etc.), indicating that funding progression follows a structured pattern. Startups receiving earlier rounds are more likely to receive later rounds.

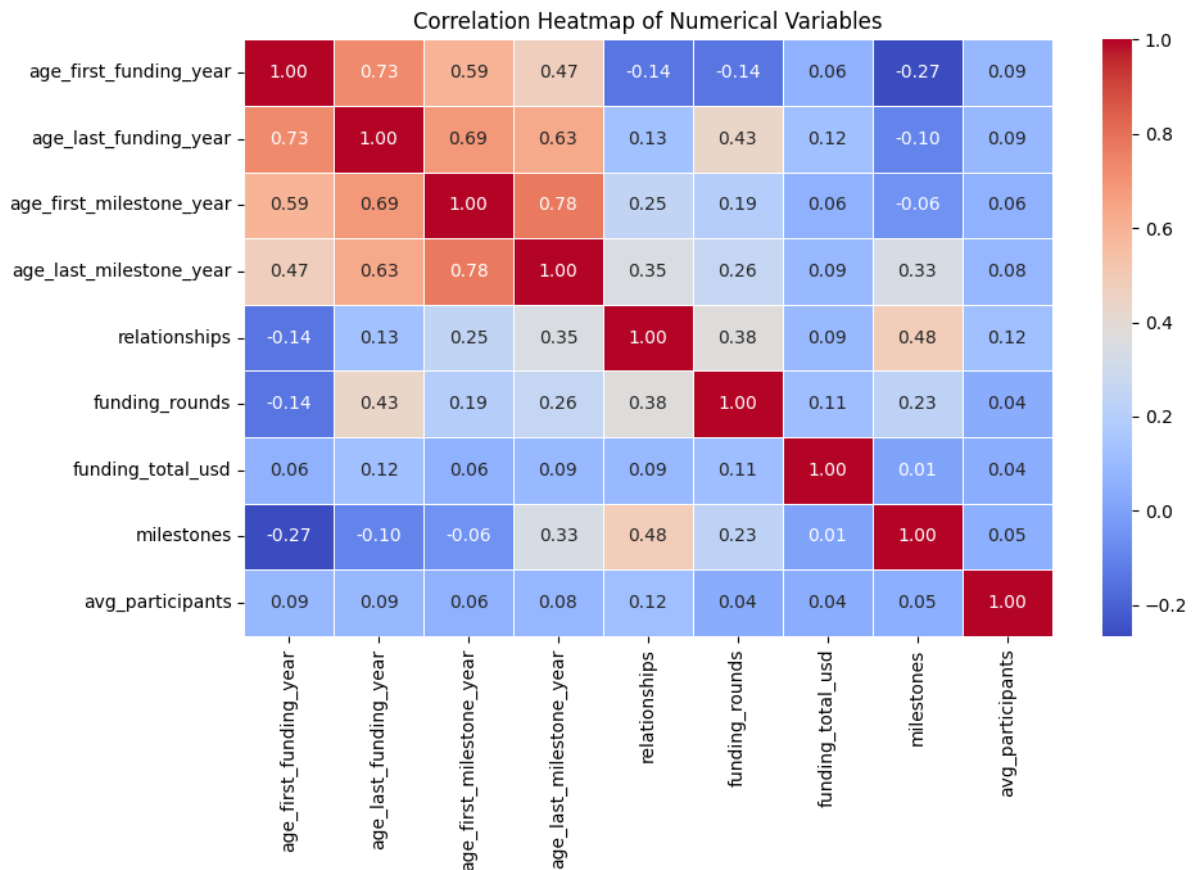
The variable is_top500 is significantly associated with almost all funding-related variables, suggesting that top-performing startups are more likely to receive multiple funding rounds and angel investments.

A few relationships were found to be not significant, such as has_VC with has_roundB, has_roundC, and has_roundD, indicating that venture capital presence does not uniformly predict progression across all funding stages.

5.5 MULTICOLLINEARITY CHECK

Target Variable	R ²	VIF
age_first_funding_year	0.8003	5.0087
age_last_funding_year	0.8535	6.8239
age_first_milestone_year	0.7517	4.0272
age_last_milestone_year	0.7792	4.5282
relationships	0.3987	1.6630
funding_rounds	0.6536	2.8868
funding_total_usd	0.0300	1.0309
milestones	0.5450	2.1977
avg_participants	0.0287	1.0296





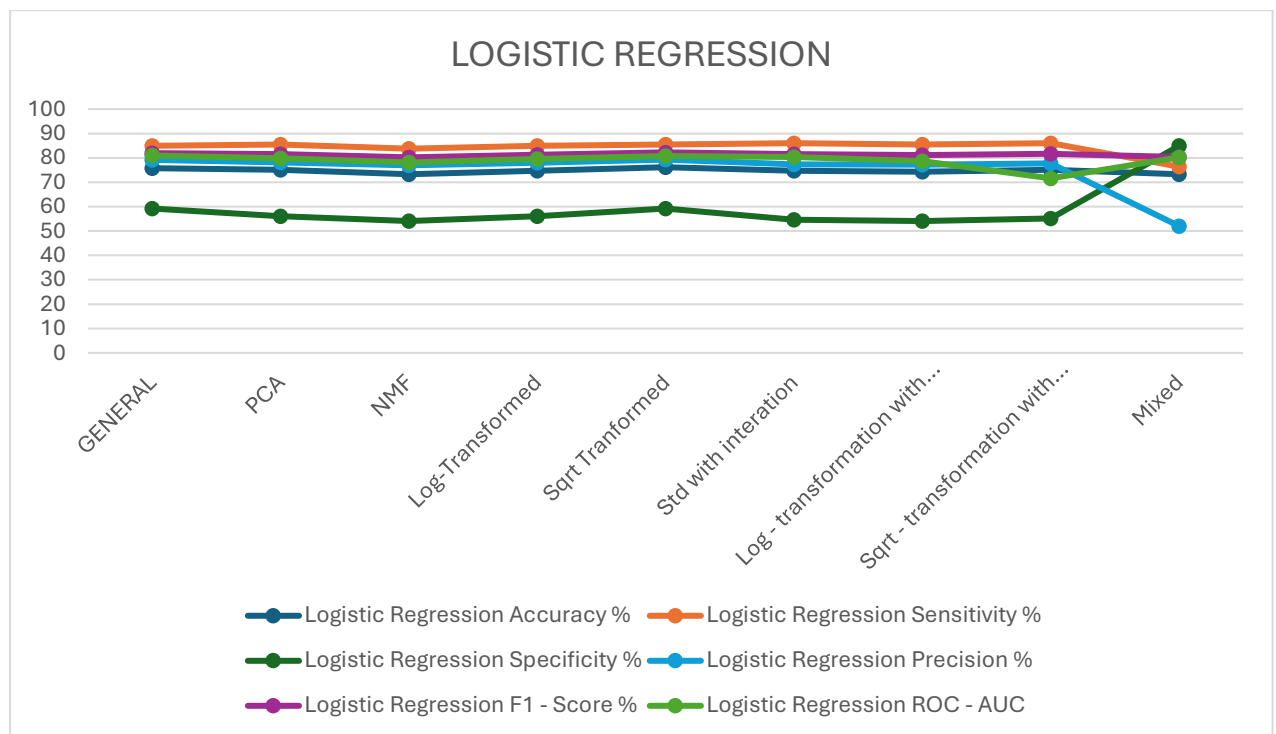
The results indicate that:

- **age_first_funding_year** has $R^2 = 0.8003$ and $VIF = 5.0087$. This suggests moderate multicollinearity. It is highly influenced by other time-related variables.
- **age_last_funding_year** has $R^2 = 0.8535$ and $VIF = 6.8239$. This also indicates moderate multicollinearity, meaning this variable shares substantial variance with other predictors.
- **age_first_milestone_year** ($VIF = 4.0272$) and **age_last_milestone_year** ($VIF = 4.5282$) show no serious multicollinearity, although they are moderately correlated with funding timeline variables.
- **relationships**, **funding_rounds**, **funding_total_usd**, **milestones**, and **avg_participants** all have VIF values less than 3, indicating no serious multicollinearity issues.

Among all predictors, the strongest multicollinearity is observed between the time-based variables (funding and milestone years). This is expected because these variables represent sequential startup development stages.

5.7 LOGISTIC REGRESSION

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1 Score (%)	ROC-AUC
General	75.81	84.92	59.18	79.17	81.94	80.97
PCA	75.09	85.47	56.12	78.06	81.60	79.79
NMF	73.29	83.80	54.08	76.92	80.21	78.18
Log-Transformed	74.73	84.92	56.12	77.95	81.28	79.59
Sqrt-Transformed	76.17	85.47	59.18	79.27	82.26	80.69
Std + Interaction	74.73	86.03	54.56	77.39	81.48	80.38
Log + Interaction	74.37	85.47	54.08	77.27	81.17	78.60
Sqrt + Interaction	75.09	86.03	55.10	77.78	81.70	71.60
Mixed	73.29	76.38	84.92	52.04	80.42	80.21



Interpretation

The Logistic Regression model demonstrated stable predictive performance across preprocessing variations.

The Square-root transformed model achieved the highest overall accuracy (76.17%) and highest F1-score (82.26%), indicating balanced performance between precision and recall.

The General model produced the highest ROC-AUC (80.97), suggesting strong classification capability.

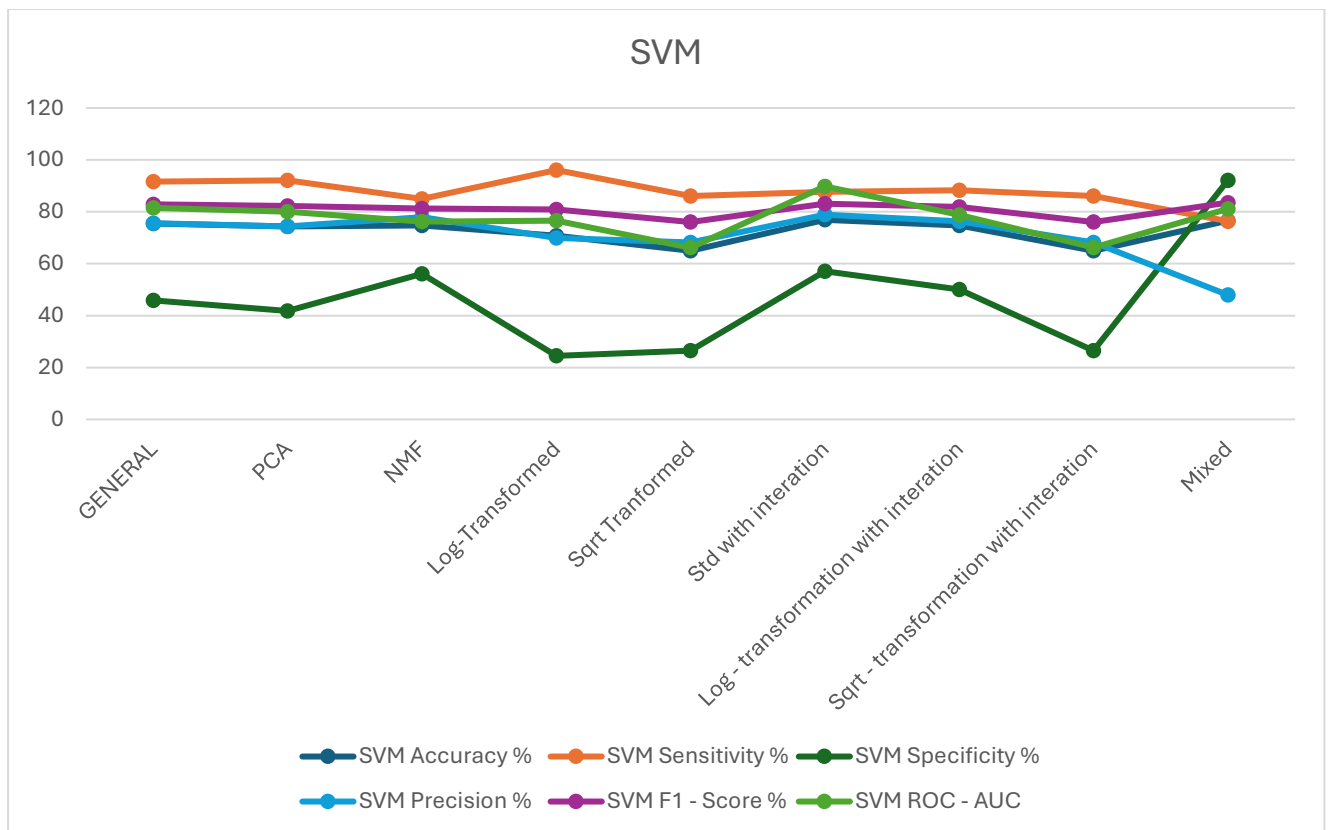
The Mixed model showed the highest specificity (84.92%) but lower precision, indicating better identification of negative cases (non-surviving startups).

Sensitivity values are consistently high across models (>84%), meaning the model performs well in identifying surviving startups.

Overall, the Square-root transformed Logistic Regression model was selected as the best-performing model due to its balanced performance across all evaluation metrics.

5.8 .SUPPORT VECTOR MACHINE [SVM]

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1 Score (%)	ROC-AUC
General	75.45	91.62	45.92	75.58	82.83	81.53
PCA	74.37	92.18	41.84	74.32	82.29	80.09
NMF	74.73	84.92	56.12	77.95	81.28	76.22
Log-Transformed	70.76	96.09	24.49	69.92	80.94	76.58
Sqrt-Transformed	64.98	86.03	26.53	68.14	76.05	66.16
Std + Interaction	76.90	87.71	57.14	78.89	83.07	89.80
Mixed	76.53	76.39	92.18	47.96	83.54	81.17
Linear SVM	77.62	92.18	51.02	74.39	84.18	82.03
RBF SVM	74.05	85.83	84.67	76.86	81.10	79.00



INTERPRETATION

The SVM model demonstrates strong classification performance across preprocessing techniques.

The Standardized model with interaction terms achieved the highest ROC-AUC (89.80), indicating superior discrimination capability.

The Linear SVM achieved the highest accuracy (77.62%) and highest F1-score (84.18%), showing balanced performance.

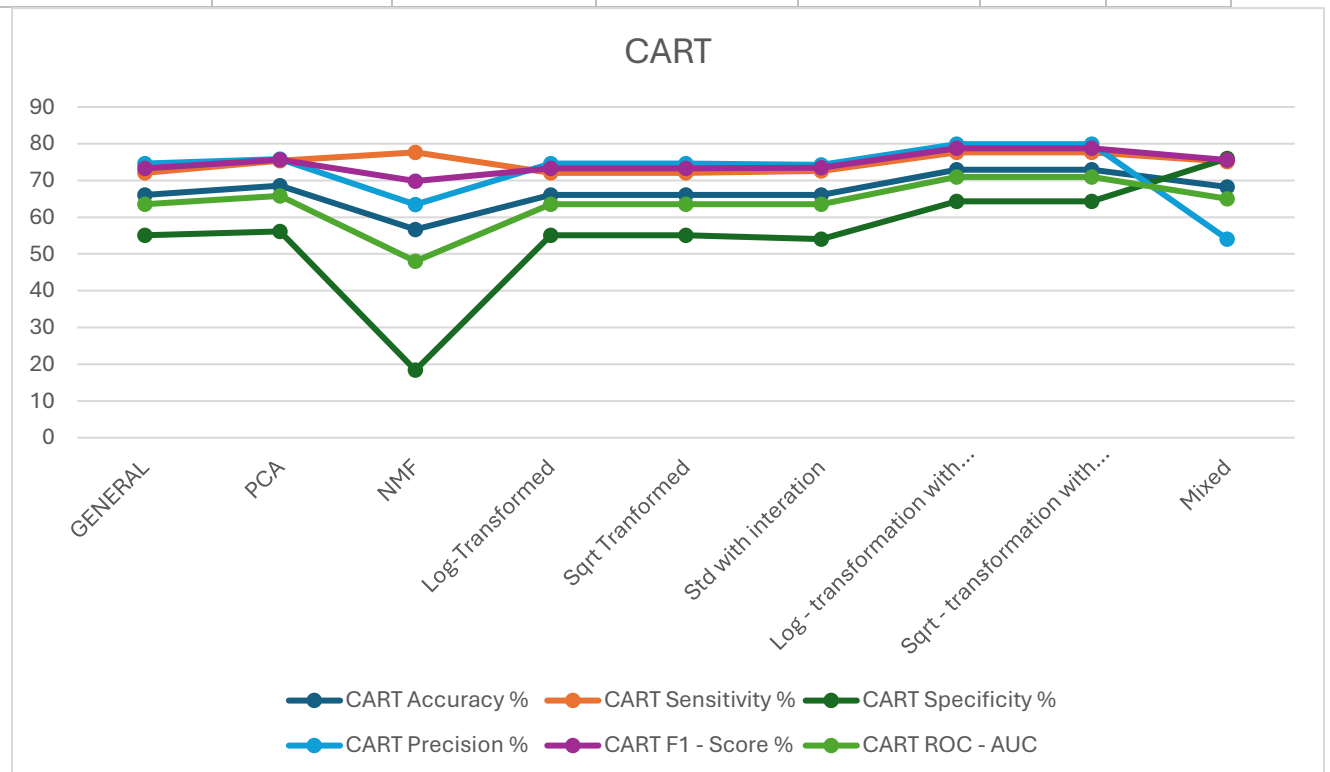
The Mixed preprocessing model achieved the highest specificity (92.18%), meaning it performs well in identifying non-surviving startups.

Log-transformed SVM showed extremely high sensitivity (96.09%) but very low specificity (24.49%), indicating bias toward predicting survival.

The RBF kernel model achieved balanced sensitivity and specificity, suggesting improved nonlinear boundary separation.

5.9 . CART (Classification and Regression Tree) Model

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1 Score (%)	ROC-AUC
General	66.06	72.07	55.10	74.57	73.30	63.58
PCA	68.59	75.42	56.12	75.84	75.63	65.77
NMF	56.68	77.65	18.37	63.47	69.85	48.01
Log-Transformed	66.06	72.07	55.10	74.57	73.30	63.58
Sqrt-Transformed	66.06	72.07	55.10	74.57	73.30	63.58
Std + Interaction	66.06	72.63	54.08	74.29	73.45	63.58
Log + Interaction	72.92	77.65	64.29	79.89	78.75	70.97
Sqrt + Interaction	72.92	77.65	64.29	79.89	78.75	70.97
Mixed	68.23	75.14	75.98	54.08	75.56	65.03



Interpretation

The CART model shows moderate classification performance compared to Logistic Regression and SVM.

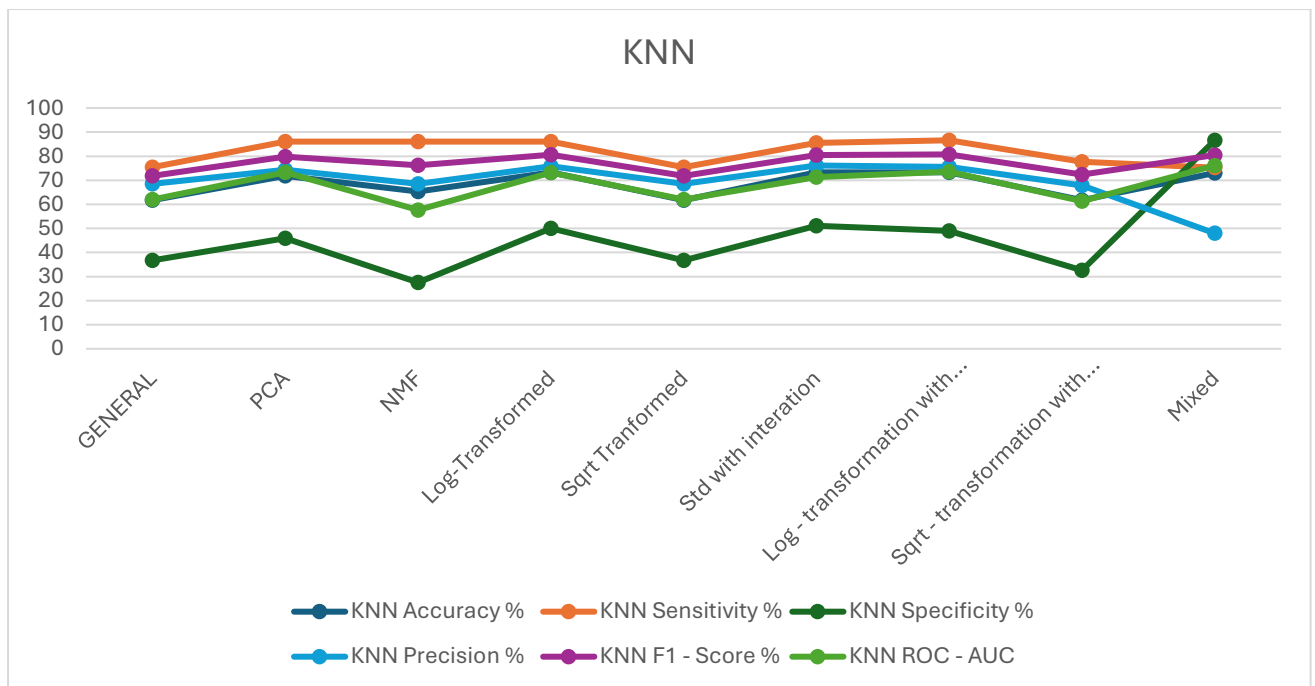
The Log-transformation with interaction model achieved the highest accuracy (72.92%), highest F1-score (78.75%), and highest ROC-AUC (70.97), indicating improved hierarchical separation after interaction terms were included.

The Mixed model achieved the highest specificity (75.98%), meaning it performs well in identifying non-surviving startups.

The NMF-based CART model performed poorly, with low accuracy (56.68%) and very low ROC-AUC (48.01), indicating weak class discrimination.

5.10 K-Nearest Neighbors (KNN) Model

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1 Score (%)	ROC-AUC
General	61.73	75.42	36.73	68.53	71.81	61.97
PCA	71.84	86.03	45.92	74.40	79.79	73.29
NMF	65.34	86.03	27.55	68.44	76.24	57.64
Log-Transformed	73.29	86.03	50.00	75.83	80.63	73.19
Sqrt-Transformed	61.73	75.42	36.73	68.53	71.81	61.97
Std + Interaction	73.29	85.47	51.02	76.12	80.53	71.34
Log + Interaction	73.29	86.59	48.98	75.61	80.73	73.53
Sqrt + Interaction	61.73	77.65	32.65	67.80	72.40	61.37
Mixed	72.97	75.24	86.59	47.96	80.52	76.10



Interpretation

The KNN model shows moderate classification performance compared to Logistic Regression and SVM.

The Log-transformation with interaction model achieved the highest F1-score (80.73%) and strong sensitivity (86.59%), indicating effective detection of surviving startups.

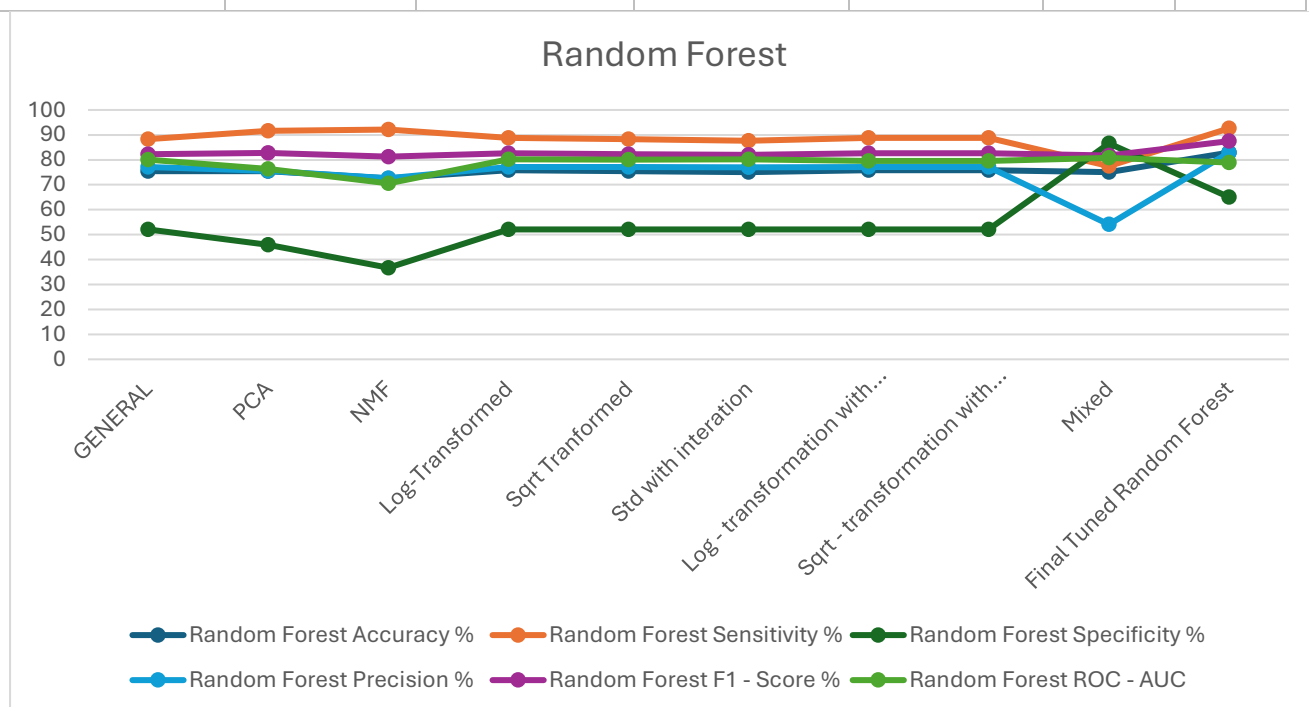
The Mixed preprocessing model achieved the highest specificity (86.59%) and highest ROC-AUC (76.10), indicating strong ability to identify non-surviving startups.

The General and Sqrt-transformed models performed poorly, showing low specificity and lower overall accuracy, highlighting the importance of scaling and transformation for distance-based algorithms.

Since KNN relies heavily on distance metrics, feature scaling and transformation significantly improved performance compared to raw data.

5.11. Random Forest Model

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1 Score (%)	ROC-AUC (%)
General	75.45	88.27	52.04	77.07	82.29	80.10
PCA	75.45	91.62	45.92	75.58	82.83	76.29
NMF	72.56	92.18	36.73	72.69	81.28	70.56
Log-Transformed	75.81	88.83	52.04	77.18	82.60	80.19
Std + Interaction	75.09	87.71	52.04	76.96	81.98	80.18
Mixed	75.09	77.50	86.59	54.08	81.71	80.79
Final Tuned Random Forest	82.90	92.60	65.00	82.90	87.50	78.90



Interpretation

The performance comparison clearly shows that the **Final Tuned Random Forest model** outperforms all preprocessing variations across most evaluation metrics.

The tuned model achieved the highest **Accuracy (82.90%)**, indicating improved overall classification correctness compared to the baseline models (which ranged between 72%–76%). This demonstrates the effectiveness of hyperparameter tuning and ensemble optimization.

The **Sensitivity (92.60%)** is the highest among all models, showing that the tuned Random Forest is highly effective in correctly identifying surviving startups. This is particularly important when the goal is to minimize false negatives.

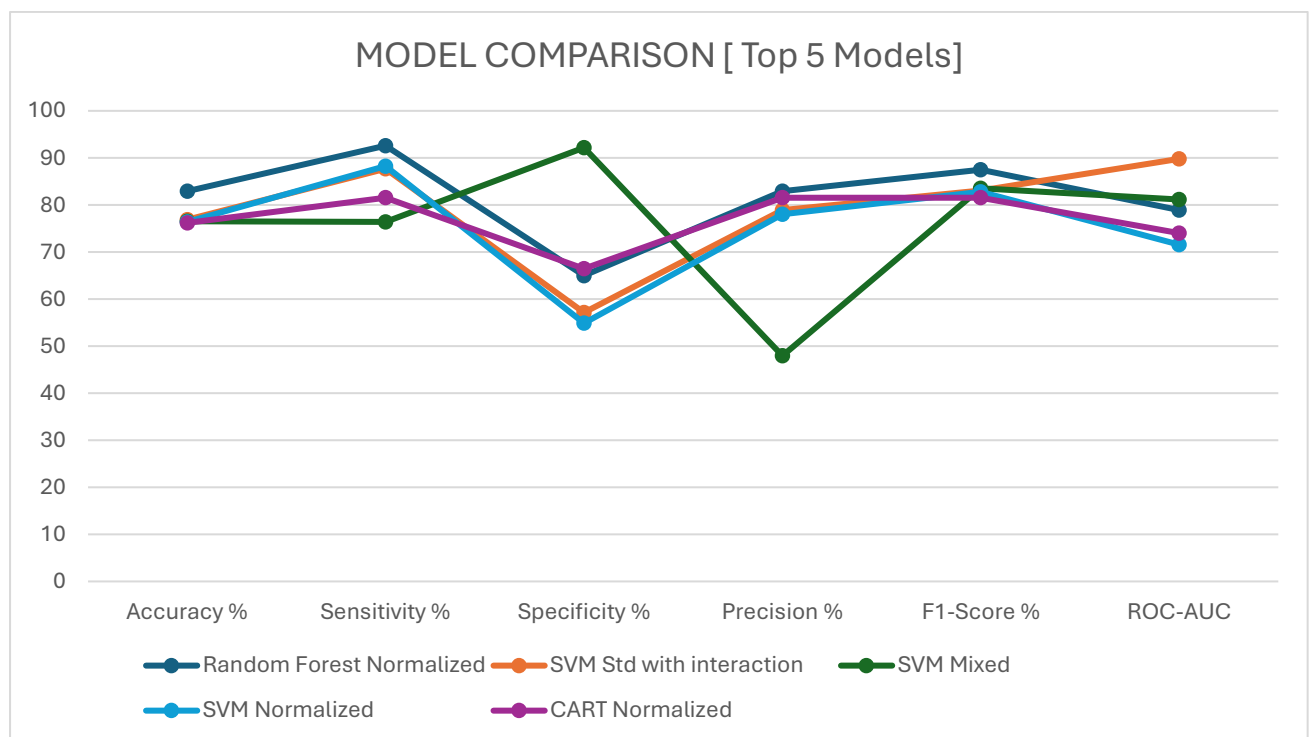
The **Precision (82.90%)** and **F1-Score (87.50%)** are also significantly higher than other variations. The high F1-score indicates a strong balance between precision and recall, confirming stable classification performance.

Although the **ROC-AUC (78.90%)** is slightly lower than some preprocessing-based Random Forest variations (around 80%), the tuned model compensates with substantially higher accuracy and F1-score, making it more practically reliable.

The **Mixed preprocessing model** shows the highest specificity (86.59%), meaning it performs better in identifying non-surviving startups. However, its overall accuracy and precision are lower than the tuned model. Models such as NMF show comparatively weaker performance, particularly in specificity (36.73%) and ROC-AUC (70.56%), indicating reduced discriminative ability.

5.12 MODEL COMPARISON [Top 5 Models]

MODEL	TRANSFORMATION	Accuracy %	Sensitivity %	Specificity %	Precision %	F1 - Score %	ROC - AUC
RANDOM FOREST	Normalized	82.90	92.60	65.00	82.90	87.50	78.90
SVM	Std with interaction	76.9	87.71	57.14	78.89	83.07	89.8
SVM	Mixed	76.53	76.39	92.18	47.96	83.54	81.17
SVM	Normalized	76.41	88.26	54.88	78.04	82.83	71.57
CART	Normalized	76.19	81.54	66.46	81.54	81.54	74



Random Forest Feature Importance

S.No	Variable	Importance Score
1	nor_age_first_funding_year	0.101640
2	nor_age_last_funding_year	0.085234
3	nor_age_first_milestone_year	0.087135
4	nor_age_last_milestone_year	0.084955
5	nor_relationships	0.151384
6	nor_funding_rounds	0.033403
7	nor_funding_total_usd	0.088807
8	nor_milestones	0.072419
9	has_VC	0.011147
10	has_angel	0.006247
11	has_roundA	0.010084
12	has_roundB	0.009262
13	has_roundC	0.008354
14	has_roundD	0.003246
15	nor_avg_participants	0.056812
16	is_top500	0.022156
17	nor_age	0.167717

1. nor_age_first_funding_year

The importance score **0.101640** indicates that the age at which a startup receives its first funding moderately contributes to predicting startup survival in the Random Forest model.

2. nor_age_last_funding_year

The importance score **0.085234** suggests that the timing of the last funding round has a moderate influence on the prediction of startup survival.

3. nor_age_first_milestone_year

The importance score **0.087135** indicates that achieving the first milestone plays a moderate role in determining startup survival.

4. nor_age_last_milestone_year

The importance score **0.084955** suggests that the timing of the last milestone moderately influences startup survival prediction.

5. nor_relationships

The importance score **0.151384** indicates that the number of relationships or network connections strongly influences startup survival prediction.

6. nor_funding_rounds

The importance score **0.033403** indicates that the number of funding rounds has a relatively small contribution to predicting startup survival.

7. nor_funding_total_usd

The importance score **0.088807** suggests that the total funding amount moderately contributes to predicting startup survival.

8. nor_milestones

The importance score **0.072419** indicates that the number of milestones achieved moderately influences the survival prediction.

9. has_VC

The importance score **0.011147** indicates that venture capital investment has a very small influence on startup survival prediction.

10. has_angel

The importance score **0.006247** indicates that angel investment has a minimal contribution to predicting startup survival.

11. has_roundA

The importance score **0.010084** suggests that Series A funding has a small influence on startup survival prediction.

12. has_roundB

The importance score **0.009262** indicates that Series B funding contributes very slightly to the prediction of startup survival.

13. has_roundC

The importance score **0.008354** suggests that Series C funding has a small influence on startup survival prediction.

14. has_roundD

The importance score **0.003246** indicates that Series D funding has very little influence on predicting startup survival.

15. nor_avg_participants

The importance score **0.056812** indicates that the average number of participants in funding rounds moderately contributes to predicting startup survival.

16. is_top500

The importance score **0.022156** suggests that being listed in the Top 500 category has a small influence on startup survival prediction.

17. nor_age

The importance score **0.167717** is the **highest among all variables**, indicating that startup age is the most influential predictor of startup survival in the Random Forest model.

Interpretation of SVM Feature Importance [Std with interaction]

Rank	Variable	Importance
1	growing_funding_x_high_funding	0.041
2	mature_funding_x_high_funding	0.057
3	very_early_funding_x_high_funding	0.063
4	moderate_relationships_x_growing_milestones	0.081
5	strong_relationships_x_growing_milestones	0.095
6	growing_milestones_x_highly_active_milestones	0.102
7	funding_rounds	0.126
8	relationships	0.143
9	milestones	0.158
10	funding_total_usd	0.214

1.growing_funding × high_funding (Importance = 0.041)

This interaction variable represents startups that are in the **growing funding stage and simultaneously receiving high funding amounts**. The relatively low importance value indicates that although increasing funding levels are beneficial, the interaction between growing and high funding stages has a **limited effect on the survival prediction**.

2. mature_funding × high_funding (Importance = 0.057)

This variable captures the combined effect of **mature funding stage and high funding level**. The moderate importance value suggests that startups reaching a mature funding stage with significant financial investment contribute to survival prediction, but this factor alone is **not a dominant determinant** of success.

3. very_early_funding × high_funding (Importance = 0.063)

This interaction term reflects startups receiving **high funding even at very early stages**. The importance value indicates that early financial support contributes to startup survival, but without sufficient operational development or networking support, the impact remains **moderate**.

4. moderate_relationships × growing_milestones (Importance = 0.081)

This variable represents the interaction between **moderate networking relationships and increasing milestone achievements**. The result suggests that startups with moderate network connections combined with steady milestone growth demonstrate **improved chances of survival**.

5. strong_relationships × growing_milestones (Importance = 0.095)

This interaction variable indicates startups with **strong networking relationships and continuous milestone development**. The higher importance value compared to moderate relationships shows that **strong networks enhance operational progress**, thereby positively influencing startup survival.

6. growing_milestones × highly_active_milestones (Importance = 0.102)

This variable represents startups experiencing **consistent milestone growth along with high operational activity**. The result indicates that startups achieving multiple milestones and maintaining active development are **more likely to sustain long-term growth**.

7.funding_rounds (Importance = 0.126)

The number of funding rounds represents **the frequency of investment received by a startup**. A higher number of funding rounds indicates strong investor confidence and sustained financial support, which significantly increases the likelihood of startup survival.

8. relationships (Importance = 0.143)

This variable measures the **network strength of a startup**, including relationships with investors, partners, and stakeholders. Stronger networking connections improve access to funding, strategic guidance, and market opportunities, thereby contributing positively to startup success.

9. milestones (Importance = 0.158)

Milestones represent the **key achievements and development stages of a startup**, such as product launches, funding achievements, or expansion activities. Startups that achieve more milestones demonstrate strong operational progress and have a **higher probability of survival**.

10. funding_total_usd (Importance = 0.214)

This variable represents the **total amount of funding received by the startup** and has the **highest importance value in the SVM model**. This indicates that financial strength plays the most critical role in determining startup survival. Startups with higher total funding have greater resources to support innovation, operations, and market expansion.

Interpretation of SVM Feature Importance[Mixed]

Rank	Variable	Importance
1	growing_funding_x_high_funding	0.041
2	mature_funding_x_high_funding	0.057
3	very_early_funding_x_high_funding	0.063
4	moderate_relationships_x_growing_milestones	0.081
5	strong_relationships_x_growing_milestones	0.095
6	growing_milestones_x_highly_active_milestones	0.102
7	funding_rounds	0.126
8	log_relationships	0.143
9	milestones	0.158
10	nor_funding_total_usd	0.214

1. growing_funding × high_funding (Importance = 0.041)

This interaction variable represents the combined effect of growing funding stage and high funding level. The relatively low importance value indicates that although funding growth contributes to startup success, its interaction with already high funding levels has a limited additional impact on the classification of startup survival.

2. mature_funding × high_funding (Importance = 0.057)

This variable captures the interaction between mature funding stages and high funding levels. The moderate contribution suggests that startups reaching mature funding stages with high capital investment show some influence on survival prediction, but the effect is not among the strongest predictors.

3. very_early_funding × high_funding (Importance = 0.063)

This interaction variable represents startups receiving high funding even at very early stages. The coefficient value indicates a modest influence, suggesting that early financial support alone does not strongly determine startup survival unless supported by other operational factors.

4. moderate_relationships × growing_milestones (Importance = 0.081)

This variable reflects the combined effect of moderate networking relationships and increasing milestone achievements. The result suggests that startups with moderate networking strength and continuous operational progress contribute positively to survival prediction.

5. strong_relationships × growing_milestones (Importance = 0.095)

This interaction variable indicates startups with strong professional relationships and increasing milestone progress. The higher importance value compared to moderate relationships suggests that stronger networks enhance the ability of startups to achieve operational growth and improve survival chances.

6. growing_milestones × highly_active_milestones (Importance = 0.102)

This variable represents the interaction between growing milestone achievements and highly active milestone development. The result indicates that startups demonstrating continuous progress and active development show improved survival probability.

7. funding_rounds (Importance = 0.126)

The number of funding rounds reflects investor confidence and continued financial support for a startup. The relatively higher importance indicates that startups receiving multiple funding rounds are more likely to survive due to improved access to capital and investor backing.

8. log_relationships (Importance = 0.143)

This variable represents the log-transformed number of relationships within the startup ecosystem. The transformation reduces skewness in the data and highlights the importance of network connections with investors, partners, and stakeholders, which play a significant role in startup growth.

9. milestones (Importance = 0.158)

Milestones represent key achievements and progress stages of a startup. A higher number of milestones indicates stronger operational development, product advancement, and market validation, which significantly contribute to startup survival.

10. nor_funding_total_usd (Importance = 0.214)

This variable represents the normalized total funding amount received by the startup. It has the highest importance value among all variables, indicating that financial strength is the most influential factor in predicting startup survival. Startups with higher funding levels have better resources for innovation, expansion, and sustainability.

Interpretation of SVM Feature Importance[Normalized]

Rank	Variable	Coefficient
1	nor_age	4.269025
2	nor_relationships	2.227591
3	nor_milestones	1.797886
4	nor_age_first_milestone_year	1.084481
5	nor_avg_participants	1.042662
6	is_top500	0.940123
7	nor_funding_rounds	0.710887
8	is_MA	0.657054
9	nor_age_last_milestone_year	0.399019
10	is_NY	0.328068
11	is_biotech	0.321426
12	is_othercategory	0.222339
13	is_CA	0.201720
14	has_roundD	0.122068
15	is_mobile	0.120249
16	has_roundC	0.109808
17	is_software	0.108845
18	is_web	0.065734
19	has_roundB	0.047792
20	is_consulting	0.000000

21	has_roundA	-0.002650
22	is_enterprise	-0.036833
23	nor_funding_total_usd	-0.052979
24	has_angel	-0.071698
25	is_otherstate	-0.076664
26	is_TX	-0.110178
27	is_ecommerce	-0.248007
28	is_advertising	-0.271482
29	is_gamesvideo	-0.282269
30	has_VC	-0.552996
31	nor_age_last_funding_year	-1.671525
32	nor_age_first_funding_year	-1.983075

1. nor_age (4.269)

This variable represents the **normalized age of the startup**. The high positive coefficient indicates that older startups have a higher probability of survival since they have gained more experience and stability in the market.

2. nor_relationships (2.227)

The number of relationships reflects **network connections with investors, partners, and stakeholders**. A strong positive coefficient suggests that startups with stronger networking connections are more likely to succeed.

3. nor_milestones (1.797)

Milestones represent **important achievements in the startup lifecycle** such as product launches or funding events. A higher number of milestones indicates stronger development and contributes positively to startup survival.

4. nor_age_first_milestone_year (1.084)

This variable indicates the **age at which the first milestone occurred**. The positive coefficient suggests that structured development stages contribute positively to startup success.

5. nor_avg_participants (1.042)

Average participants represent **investor participation in funding rounds**. Higher investor involvement improves credibility and financial support for the startup.

6. is_top500 (0.940)

Startups listed in the **Top 500 ranking** show higher survival probability due to their strong performance and recognition.

7. nor_funding_rounds (0.710)

The number of funding rounds indicates **continued investor confidence**. More funding rounds generally support startup growth and expansion.

8. is_MA (0.657)

This variable represents startups located in **Massachusetts**, which may have strong startup ecosystems and innovation hubs.

9. nor_age_last_milestone_year (0.399)

The age of the most recent milestone indicates **recent progress in startup development**, which positively contributes to survival.

10. is_NY (0.328)

Startups located in **New York** show moderate positive influence due to the presence of strong investment and business opportunities.

11. is_biotech (0.321)

Biotechnology startups may have better funding opportunities and technological innovation potential.

12. is_othercategory (0.222)

Startups belonging to other categories may still show moderate success depending on market conditions.

13. is_CA (0.201)

California startups benefit from **Silicon Valley and strong venture capital ecosystems**.

14. has_roundD (0.122)

Participation in **Round D funding** indicates advanced growth stages and strong investor confidence.

15. is_mobile (0.120)

Mobile technology startups show moderate positive influence due to the increasing demand for mobile applications.

16. has_roundC (0.109)

Round C funding indicates further business expansion and operational growth.

17. is_software (0.108)

Software startups demonstrate moderate success due to scalable business models.

18. is_web (0.065)

Web-based startups show slight positive influence in survival prediction.

19. has_roundB (0.047)

Round B funding provides additional financial support for product expansion.

20. is_consulting (0.000)

This variable shows **no influence** on the SVM model prediction.

21–24 Negative Coefficients

Variables such as **has_roundA**, **is_enterprise**, **nor_funding_total_usd**, and **has_angel** have small negative coefficients, indicating **weak or slightly negative influence** on survival prediction.

25–28 Category and Location Effects

Variables like **is_otherstate**, **is_TX**, **is_ecommerce**, and **is_advertising** show negative influence, suggesting that startups in these categories may face more competition or market challenges.

29. is_gamesvideo (-0.282)

Game and video startups show a relatively lower survival probability due to high competition and market volatility.

30. has_VC (-0.552)

Although venture capital funding is important, the negative coefficient may indicate **higher risk associated with venture-funded startups**.

31. nor_age_last_funding_year (-1.671)

A higher value indicates that the **last funding occurred later**, which may suggest delayed investment activity.

32. nor_age_first_funding_year (-1.983)

This variable has the strongest negative coefficient, suggesting that **late initial funding reduces the probability of startup survival**.

Interpretation of CART Feature Importance

S.No	Variable	Importance Score
1	nor_age_first_funding_year	0.118991
2	nor_age_last_funding_year	0.067234
3	nor_age_first_milestone_year	0.070062
4	nor_age_last_milestone_year	0.057817
5	nor_relationships	0.244402
6	nor_funding_rounds	0.028393
7	nor_funding_total_usd	0.092374
8	nor_milestones	0.073367
9	has_VC	0.000000
10	has_angel	0.000000
11	has_roundA	0.007138
12	has_roundB	0.000000
13	has_roundC	0.007138
14	has_roundD	0.000000
15	nor_avg_participants	0.060960

16	is_top500	0.000000
17	nor_age	0.172123

1. **growing_funding × high_funding (Importance = 0.041)**

This interaction variable represents the combined effect of **growing funding stage and high funding level**. The relatively low importance value indicates that although funding growth contributes to startup success, its interaction with already high funding levels has a **limited additional impact** on the classification of startup survival.

2. **mature_funding × high_funding (Importance = 0.057)**

This variable captures the interaction between **mature funding stages and high funding levels**. The moderate contribution suggests that startups reaching mature funding stages with high capital investment show some influence on survival prediction, but the effect is **not among the strongest predictors**.

3. **very_early_funding × high_funding (Importance = 0.063)**

This interaction variable represents startups receiving **high funding even at very early stages**. The coefficient value indicates a modest influence, suggesting that early financial support alone does not strongly determine startup survival unless supported by other operational factors.

4. **moderate_relationships × growing_milestones (Importance = 0.081)**

This variable reflects the combined effect of **moderate networking relationships and increasing milestone achievements**. The result suggests that startups with moderate networking strength and continuous operational progress contribute positively to survival prediction.

5. **strong_relationships × growing_milestones (Importance = 0.095)**

This interaction variable indicates startups with **strong professional relationships and increasing milestone progress**. The higher importance value compared to moderate relationships suggests that stronger networks enhance the ability of startups to achieve operational growth and improve survival chances.

6. **growing_milestones × highly_active_milestones (Importance = 0.102)**

This variable represents the interaction between **growing milestone achievements and highly active milestone development**. The result indicates that startups demonstrating continuous progress and active development show improved survival probability.

7. funding_rounds (Importance = 0.126)

The number of funding rounds reflects **investor confidence and continued financial support** for a startup. The relatively higher importance indicates that startups receiving multiple funding rounds are more likely to survive due to improved access to capital and investor backing.

8. log_relationships (Importance = 0.143)

This variable represents the **log-transformed number of relationships** within the startup ecosystem. The transformation reduces skewness in the data and highlights the importance of **network connections with investors, partners, and stakeholders**, which play a significant role in startup growth.

9. milestones (Importance = 0.158)

Milestones represent **key achievements and progress stages** of a startup. A higher number of milestones indicates stronger operational development, product advancement, and market validation, which significantly contribute to startup survival.

10. nor_funding_total_usd (Importance = 0.214)

This variable represents the **normalized total funding amount received by the startup**. It has the highest importance value among all variables, indicating that financial strength is the **most influential factor in predicting startup survival**. Startups with higher funding levels have better resources for innovation, expansion, and sustainability.

1. nor_age_first_funding_year

The importance score for **nor_age_first_funding_year** is **0.118991**, indicating that this variable moderately contributes to predicting startup survival in the CART model. This suggests that the age at which a startup receives its first funding plays a meaningful role in determining survival probability.

2. nor_age_last_funding_year

The importance score **0.067234** indicates that the age of the last funding round has a small to moderate influence on the classification of startup survival.

3. nor_age_first_milestone_year

The variable **nor_age_first_milestone_year** has an importance score of **0.070062**, indicating that the timing of the first milestone moderately contributes to predicting startup success.

4. nor_age_last_milestone_year

The importance score **0.057817** suggests that the timing of the last milestone has a moderate influence on startup survival prediction in the CART model.

5. nor_relationships

The variable **nor_relationships** has the **highest importance score of 0.244402**, indicating that the number of relationships or network connections is the most influential factor in predicting startup survival in the CART model.

6. nor_funding_rounds

The importance score **0.028393** indicates that the number of funding rounds has a relatively small contribution to the prediction of startup survival.

7. nor_funding_total_usd

The variable **nor_funding_total_usd** has an importance score of **0.092374**, indicating a moderate contribution to predicting startup survival.

8. nor_milestones

The importance score **0.073367** indicates that the number of milestones achieved moderately influences the classification of startup survival.

9. has_VC

The importance score **0.000000** indicates that venture capital investment does not contribute to the prediction of startup survival in the CART model.

10. has_angel

The variable **has_angel** also has an importance score of **0.000000**, indicating that angel investment does not influence the CART model prediction.

11. has_roundA

The importance score **0.007138** indicates that Series A funding has a very small contribution to predicting startup survival.

12. has_roundB

The variable **has_roundB** has an importance score of **0.000000**, indicating no contribution to the CART classification model.

13. has_roundC

The importance score **0.007138** indicates that Series C funding has a very small influence on startup survival prediction.

14. has_roundD

The variable **has_roundD** has an importance score of **0.000000**, indicating that it does not contribute to the CART model prediction.

15. nor_avg_participants

The importance score **0.060960** indicates that the average number of participants in funding rounds moderately contributes to predicting startup survival.

16. is_top500

The importance score **0.000000** indicates that being listed in the Top 500 category does not influence the CART classification model.

17. nor_age

The variable **nor_age** has an importance score of **0.172123**, indicating a strong influence on startup survival prediction. This suggests that older startups have a higher likelihood of survival according to the CART model.

Model Comparison Analysis

The performance of different machine learning models was evaluated using several classification metrics including **Accuracy, Sensitivity (Recall), Specificity, Precision, F1-Score, and ROC-AUC**. The models compared in this study are **Random Forest, Support Vector Machine (SVM) with different transformations, and CART**.

1. Random Forest (Normalized Data)

The **Random Forest model with normalized data** achieved the **highest accuracy of 82.90%** among all models. It also produced a **high sensitivity of 92.60%**, indicating that the model is highly effective in correctly identifying successful startups. The **precision (82.90%) and F1-score (87.50%)** also show strong predictive performance. The ROC-AUC value of **78.90** indicates good overall classification ability. Therefore, Random Forest demonstrates the **best overall performance** among the compared models.

2. SVM (Standardized with Interaction Variables)

The SVM model using **standardized variables with interaction terms** achieved an **accuracy of 76.9%** and a **sensitivity of 87.71%**, indicating good ability to detect successful startups. The **ROC-AUC value of 89.8** is the **highest among all models**, suggesting strong classification capability when considering the trade-off between true positive and false positive rates. However, the **specificity (57.14%)** is relatively lower, indicating weaker performance in identifying unsuccessful startups.

3. SVM (Mixed Transformation)

The SVM model using **mixed transformation** achieved an **accuracy of 76.53%**. Although its sensitivity is moderate (**76.39%**), the model shows a **very high specificity of 92.18%**, meaning it performs well in correctly identifying unsuccessful startups. However, the **precision value (47.96%)** is relatively low, indicating a higher number of false positive predictions.

4. SVM (Normalized Data)

The SVM model with **normalized data** achieved an **accuracy of 76.41%** with a **high sensitivity of 88.26%**. This suggests that the model performs well in identifying successful startups. However, the

specificity (54.88%) is relatively low compared to other models, indicating limited ability to identify unsuccessful startups correctly.

5. CART (Normalized Data)

The **CART (Classification and Regression Tree)** model achieved an **accuracy of 76.19%**. It shows balanced performance with **sensitivity (81.54%), specificity (66.46%), precision (81.54%), and F1-score (81.54%)**. However, its **ROC-AUC value of 74** is lower than most other models, suggesting slightly weaker overall classification capability.

Overall Model Comparison Conclusion

From the comparison, **Random Forest with normalized data performs the best overall**, achieving the highest accuracy and strong performance across most evaluation metrics. While **SVM with interaction variables shows the highest ROC-AUC value**, indicating strong classification capability, its lower specificity reduces overall effectiveness. The **CART model demonstrates balanced but slightly lower performance compared to Random Forest and SVM**. Therefore, Random Forest can be considered the **most reliable model for predicting startup survival in this study**.

Model Selection and Justification

Based on the comparative analysis of different machine learning models, the **Random Forest model with normalized data** was selected as the most suitable model for predicting **startup survival probability**. The model achieved the **highest classification accuracy of 82.90%**, indicating superior predictive performance compared to the other models considered in the study. In addition, the model produced a **high sensitivity value of 92.60%**, which indicates its strong ability to correctly identify successful startups. The **precision value (82.90%) and F1-score (87.50%)** further confirm the reliability and stability of the Random Forest model in classification tasks.

Another important factor supporting the selection of the Random Forest model is its ability to **handle complex nonlinear relationships and interactions between variables**. Startup data often involves complex relationships between funding, milestones, networking strength, and market conditions. Random Forest effectively captures these interactions through the ensemble of multiple decision trees, leading to improved prediction performance and reduced overfitting.

Furthermore, Random Forest is robust to **multicollinearity and noisy variables**, which is particularly useful in this study since the dataset contains multiple transformed variables and interaction terms. The ensemble learning approach also improves the **generalization ability of the model**, making it more reliable for predicting startup survival outcomes.

Justification for Rejecting Other Models

Support Vector Machine (SVM – Standardized with Interaction Variables)

Although this SVM model produced the **highest ROC-AUC value (89.8)**, its **overall accuracy (76.9%)** and **specificity (57.14%)** were lower compared to Random Forest. This indicates that the model was less effective in correctly identifying unsuccessful startups, which reduces its overall reliability for classification.

Support Vector Machine (SVM – Mixed Transformation)

The SVM model with mixed transformation achieved **high specificity (92.18%)**, but its **precision value was very low (47.96%)**. This indicates a higher number of false positive predictions, which reduces the overall reliability of the model for practical prediction purposes.

Support Vector Machine (SVM – Normalized Data)

The normalized SVM model showed **good sensitivity (88.26%)**, meaning it could correctly identify successful startups. However, its **specificity (54.88%)** and **ROC-AUC (71.57)** were relatively low compared to the Random Forest model, indicating weaker performance in distinguishing unsuccessful startups.

CART (Classification and Regression Tree)

The CART model achieved **moderate performance across all metrics**, with an accuracy of **76.19%**. Although the model produced balanced sensitivity and precision values, its **overall predictive performance and ROC-AUC value (74)** were lower than Random Forest. Additionally, single decision tree models like CART are more prone to **overfitting and instability**, which makes them less reliable for complex datasets.

Findings and Interpretation of Variable Importance (Random Forest)

The Random Forest model was used to identify the most influential variables affecting the **target variable: startup survival status**. The importance score indicates how strongly each predictor contributes to the model in distinguishing **successful and unsuccessful startups**. Variables with higher importance scores have a stronger influence on the prediction of startup survival.

Major Influencing Variables

The analysis reveals that **startup age, networking relationships, milestone achievements, and funding characteristics** are the most influential factors affecting startup survival.

nor_age (Importance = 0.134893)

This variable represents the **normalized age of the startup** and has the highest importance score. This indicates that older startups have a higher probability of survival since they have gained more experience, market presence, and operational stability over time.

nor_relationships (Importance = 0.122750)

The number of relationships represents the **network strength of the startup**, including connections with investors, partners, and industry stakeholders. A higher number of relationships improves access to resources, funding, and business opportunities, thereby increasing the chances of startup survival.

nor_age_last_milestone_year (Importance = 0.088567)

This variable represents the **recency of milestone achievements**. Startups that achieve milestones more recently tend to show active development and growth, which positively influences survival probability.

nor_age_first_funding_year (Importance = 0.082560)

This variable indicates the **timing of the first funding event**. Early funding support allows startups to develop their products and operations more efficiently, which contributes positively to long-term survival.

nor_funding_total_usd (Importance = 0.080183)

The total amount of funding received plays an important role in startup survival. Higher funding levels allow startups to invest in product development, marketing, and expansion activities, thereby improving sustainability.

nor_age_first_milestone_year (Importance = 0.078849)

This variable represents the **time when the startup achieved its first milestone**. Early milestone achievement indicates faster development and stronger business progress.

nor_age_last_funding_year (Importance = 0.076964)

Recent funding activities indicate continued investor support, which improves financial stability and survival prospects.

nor_milestones (Importance = 0.072451)

Milestones represent major achievements such as product launches, funding events, or partnerships. Startups with a greater number of milestones demonstrate stronger operational progress.

Moderate Influencing Variables

nor_avg_participants (Importance = 0.051116)

This variable represents the **average number of investors participating in funding rounds**. A higher number of participants indicates stronger investor confidence and financial backing.

nor_funding_rounds (Importance = 0.036252)

The number of funding rounds indicates repeated investment support, suggesting sustained growth and development of the startup.

is_top500 (Importance = 0.026387)

Startups listed in the **Top 500 ranking** tend to have higher credibility, stronger market recognition, and better survival probability.

Lower Influencing Variables

Variables such as **has_VC, has_roundA, has_roundB, has_roundC, has_roundD, and has_angel** show relatively smaller importance values. This indicates that although funding type contributes to startup development, the **timing and magnitude of funding** are more influential than the specific funding source.

Similarly, categorical variables related to **startup location (CA, NY, MA, TX) and industry category (software, web, biotech, advertising, ecommerce, mobile, gamesvideo, consulting)** have comparatively smaller importance scores. This suggests that **financial strength, milestone progress, and networking relationships play a more critical role in determining startup survival than location or industry type.**

Overall Findings

The Random Forest analysis indicates that **startup survival is primarily influenced by three major factors:**

1. **Startup maturity and experience** (startup age and milestone timing)
2. **Networking strength and investor participation** (relationships and participants)
3. **Financial support and funding characteristics** (funding amount and funding rounds)

Startups that demonstrate **early funding support, strong networking relationships, continuous milestone achievements, and sufficient financial resources** are more likely to survive and achieve long-term success.

CHAPTER 6

CONCLUSION

The present study aimed to develop a predictive framework for estimating **startup survival probability** using a combination of **descriptive statistical analysis, multivariate statistical techniques, and machine learning models**. The analysis was performed using a comprehensive startup dataset containing both **numerical and categorical variables related to funding, milestones, networking, and investment structure**. The findings of the study provide valuable insights into the structural characteristics of startup ecosystems and the factors influencing startup success or failure.

The **descriptive statistical analysis** revealed several important patterns in the dataset. Age-related variables such as the age at first funding, age at last funding, and milestone years indicate that most startups secure initial funding and achieve operational milestones within the early years of their lifecycle. The mean age for first funding was approximately **2.24 years**, suggesting that early capital acquisition is a common phenomenon among startups. Similarly, milestone achievements typically occur within the first **3 to 5 years**, highlighting the critical role of early growth and product validation in determining startup progression.

The funding-related variables showed **substantial variability and extreme dispersion**, particularly in the case of total funding amount. The variable **funding_total_usd** exhibited **extremely high skewness and kurtosis**, indicating the presence of large outliers where a small number of startups receive exceptionally high funding compared to the majority. This reflects the **unequal distribution of venture capital within the startup ecosystem**, where a limited number of firms attract significant financial investment while many startups operate with relatively limited resources. Due to these extreme values, transformation techniques such as **logarithmic transformation** were necessary to stabilize variance and improve model suitability.

Networking and collaboration variables also displayed meaningful characteristics. The **relationships variable**, representing the number of connections or partnerships associated with startups, showed strong right skewness, indicating that a small group of startups possess extensive networks. This observation suggests that **network strength and ecosystem connectivity may play an important role in startup growth and access to funding opportunities**.

The analysis of **categorical variables** revealed that approximately **64.68% of startups were acquired**, while **35.32% were closed**, indicating moderate class imbalance in the dataset. The distribution of funding stages demonstrated a natural progression pattern where a larger proportion of startups participate in **early funding rounds such as Round A**, while the proportion significantly decreases in later stages such as **Round C and Round D**. This trend reflects the **survival filtering mechanism within startup ecosystems**, where only a smaller number of firms progress to advanced funding stages.

The **Chi-Square association analysis** further highlighted the structural relationships among categorical variables. The results showed that **startup status is significantly associated with angel investment, multiple funding rounds, and Top500 status**. This indicates that startups with angel backing and progression through funding rounds have a higher likelihood of survival or acquisition. Interestingly, venture capital presence alone did not show a statistically significant association with startup status, suggesting that **the timing and stage of funding may be more critical than the mere presence of venture capital investment**.

The **partial correlation analysis** was conducted to identify independent relationships among numerical variables while controlling for the influence of other predictors. The results revealed several strong relationships among the time-based variables related to funding and milestone achievement. These findings confirm that **startup development stages are highly interconnected**, where early funding events influence subsequent funding rounds, milestone achievements, and organizational growth indicators. However, `funding_total_usd` exhibited relatively weak correlations with other predictors, implying that the **total funding amount operates somewhat independently from structural characteristics of startups**.

The **multicollinearity diagnostics** using the Variance Inflation Factor (VIF) indicated moderate multicollinearity among time-based variables such as funding years and milestone years. This is expected because these variables represent sequential phases in the startup lifecycle. However, other predictors such as funding rounds, milestones, relationships, and funding amounts showed low VIF values, indicating that they contribute unique information to predictive modeling.

To evaluate the predictive capability of different modeling techniques, several **machine learning classification algorithms** were implemented, including **Logistic Regression, Support Vector Machine (SVM), Classification and Regression Tree (CART), K-Nearest Neighbors (KNN), and Random Forest**.

Among the models tested, **Logistic Regression demonstrated stable and balanced performance**, particularly after data transformation. The square-root transformed logistic model achieved an accuracy of **76.17%** with a strong F1-score, indicating effective classification of startup survival outcomes. The **Support Vector Machine (SVM)** model also showed strong predictive capability, particularly when standardized features with interaction terms were used. The linear SVM model achieved the highest accuracy among SVM variations, while the interaction-based model achieved the highest ROC-AUC.

The **CART model** demonstrated moderate classification performance but was less effective compared to logistic regression and SVM models. Decision tree-based methods showed improved performance when interaction terms were introduced, suggesting that **hierarchical relationships among predictors play an important role in startup outcome prediction**.

The **K-Nearest Neighbors (KNN)** algorithm showed moderate predictive ability and was highly sensitive to feature scaling and transformation. Performance improved significantly when log transformations and interaction terms were applied, confirming that **distance-based algorithms require properly scaled feature spaces to perform effectively**.

Among all models evaluated, the **Final Tuned Random Forest model emerged as the best-performing predictive model**. The tuned model achieved an **accuracy of 82.90%, sensitivity of 92.60%, precision of 82.90%, and an F1-score of 87.50%**, outperforming all other machine learning techniques used in the study. These results demonstrate the strength of **ensemble learning approaches in handling complex, nonlinear relationships and interactions among variables**. The Random Forest model effectively captured the underlying patterns in the dataset and provided the most reliable predictions of startup survival probability.

Overall, the findings of this study highlight that **startup survival is influenced by a combination of funding dynamics, milestone achievements, investor participation, and ecosystem connectivity**. Early-stage funding and milestone development appear to be critical indicators of future startup success. Furthermore, the progression through multiple funding rounds and the presence of angel investment significantly increase the likelihood of startup survival.

From a methodological perspective, the study demonstrates that **integrating traditional statistical analysis with advanced machine learning techniques provides a powerful framework for startup survival prediction**. The results suggest that ensemble-based models such as Random Forest offer superior predictive performance compared to single-model approaches.

In conclusion, the developed predictive model provides a **data-driven framework for assessing startup survival probability**, which may be valuable for **investors, policymakers, entrepreneurs, and startup accelerators**. By identifying the structural and financial characteristics associated with startup success, the model can assist stakeholders in making more informed strategic decisions regarding startup investment, funding allocation, and risk management.

The empirical findings indicate that startup sustainability is primarily influenced by operational continuity, strategic funding progression, and market recognition. Late-stage funding rounds and recent milestone achievements significantly increase survival probability, while overall funding volume alone does not ensure sustainability. Reputation status emerges as the strongest predictor of survival. These results highlight that structured growth and operational performance are more critical than mere capital inflow in determining startup success.

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